

Stock-AI: AI Integrated Stock Trading Platform

Spring 2026

COM S 4020 - UG_HF_03

<https://seniord.cs.iastate.edu/2026-March-14/>

Client - Dr. Hung Phan

Instructor - Dr. Simanta Mitra

Teaching Assistant - Haniyeah Fekrimandi

Team Members:

Andrew Butler - Backend development, client communication

Matthew He - Full Stack development

Ethan Roe - Frontend development

Aaron Trelstad - Full-Stack development, AI integration

Problem

Stock trading is a difficult skill to learn. The stock market is constantly changing and evolving, and gaining intuition on smart investments is time consuming and difficult to test without making real investments. Learning platforms exist, but they are expensive services that add on to the already financially risky task of trading.

Solution

Our platform, Stock-AI, aims to help beginners learn stock trading completely for free with no risk. It accomplishes this in two ways: first, the Stock-AI platform doesn't charge users. There is no subscription or service cost to the user, and all trades are simulated to eliminate the risk of financial loss. Second, the platform includes AI integration to allow the user to ask questions and get summaries of relevant information, as well as to get recommendations on any given stock.

Requirements

Our team was initially tasked with identifying important features from existing platforms such as BunnyQuant and RobinHood. We created an initial proposed list of features which then guided our development of Stock-AI. We compiled and gave this list to Dr. Phan, who gave us the green light on them. While the final Stock-AI application has many of the same ideas, it is not a complete implementation of the initial featurelist; as a team, we continually gained domain knowledge and evolved what we included, given the approval of Dr. Phan on new features. The initial feature document includes user security, portfolios, mock trading, and stock history viewing, but notably does not include the leagues feature (described below). The original proposed featurelist document is appended at the end of this report.

Major Stock-AI elements

Stock-AI has 3 main pages that users interact with: the Dashboard, User Portfolios, and Leagues.

1. Dashboard:

The Dashboard serves as the primary way that users get information about the stock market. From here, they can look up any ticker, view its price history, see linked news articles about that company, as well as ask the AI assistant questions and get a recommendation for buy/sell/hold with associated reasoning. Trades can also be performed here, although they must be linked through a singular portfolio the user has. See **Figure 1** below for a view of the dashboard.

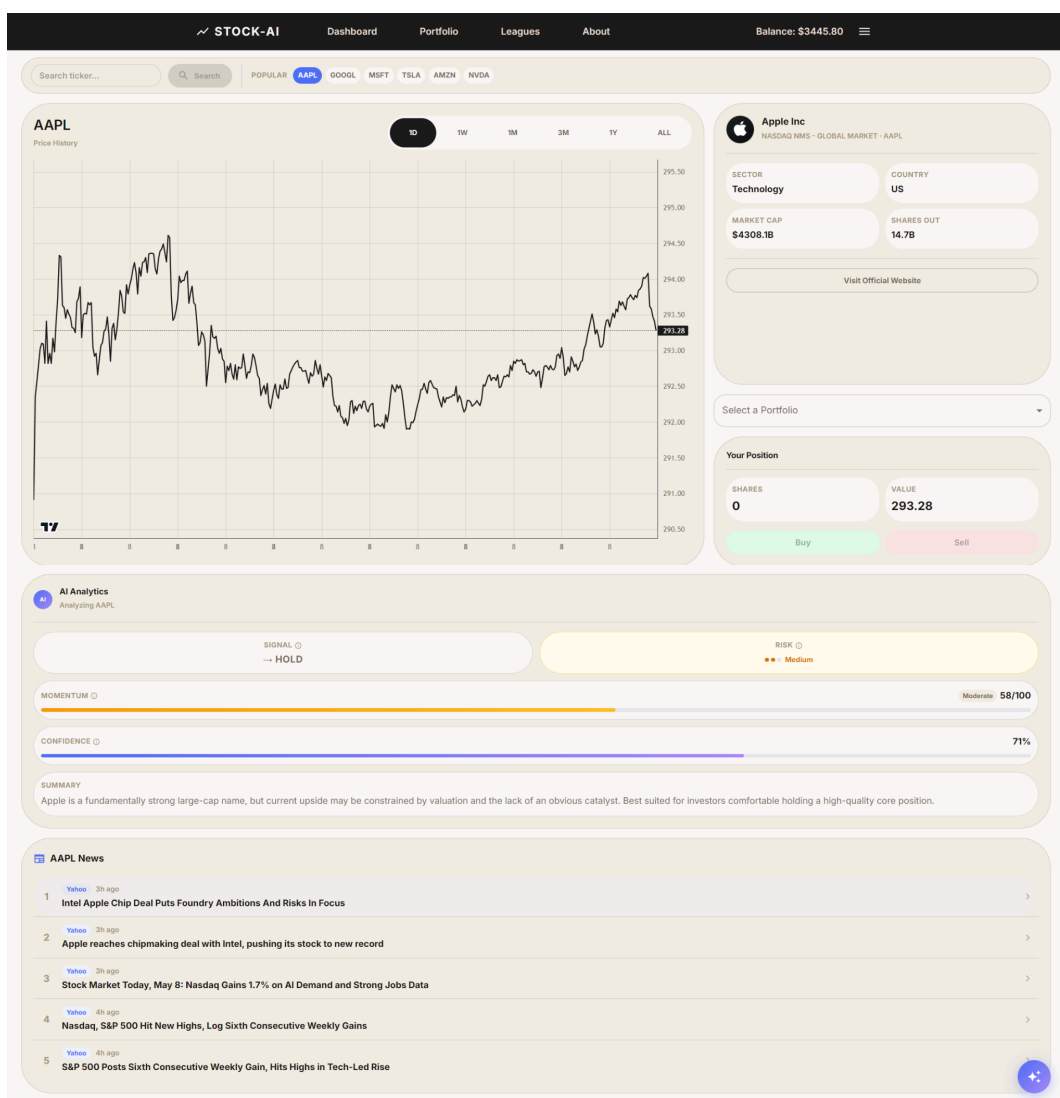


Figure 1: Dashboard page of Stock-AI platform. Note the price graph in the upper left, company information on the upper right, AI analytics in the middle, and relevant news at the bottom.

2. Portfolios:

The Portfolios view allows a user to create, manage, and view their portfolios, including any held positions they contain. Viewing positions gives their current price per share, the average cost basis of the user's owned shares, and their net gain/loss at the current market price. The user can choose to buy/sell from here as well, which will automatically lead them to the Dashboard with that given stock. **Figure 2** on the next page illustrates this view.

Portfolios also have a performance view, which enables a user to see their portfolio value over time. The graph view allows selection of time period as well as the inclusion or exclusion of any held positions. The performance view also shows trade history, stock allocation, and specific value numbers for the portfolio. See **Figure 3** on the next page for an example of the portfolio performance view.

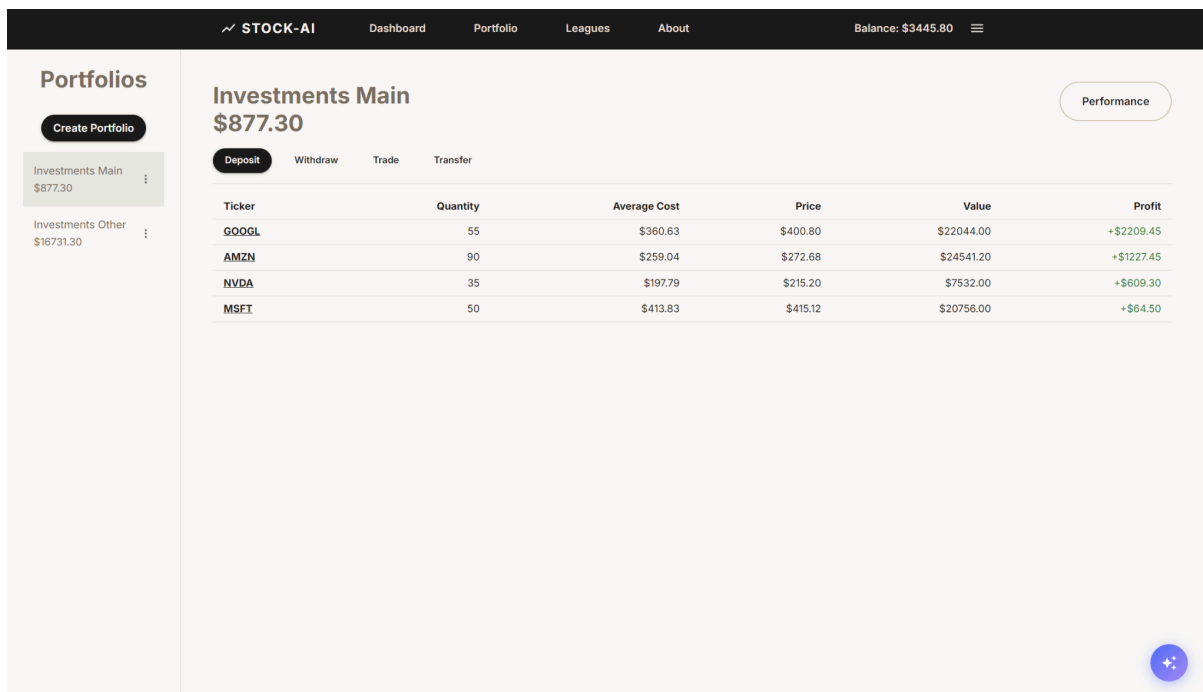


Figure 2: Portfolios view. Note the list of portfolios on the left, with position information (quantity, cost basis, current price, total current value, and net gain/loss) displayed on the right two-thirds of the screen.

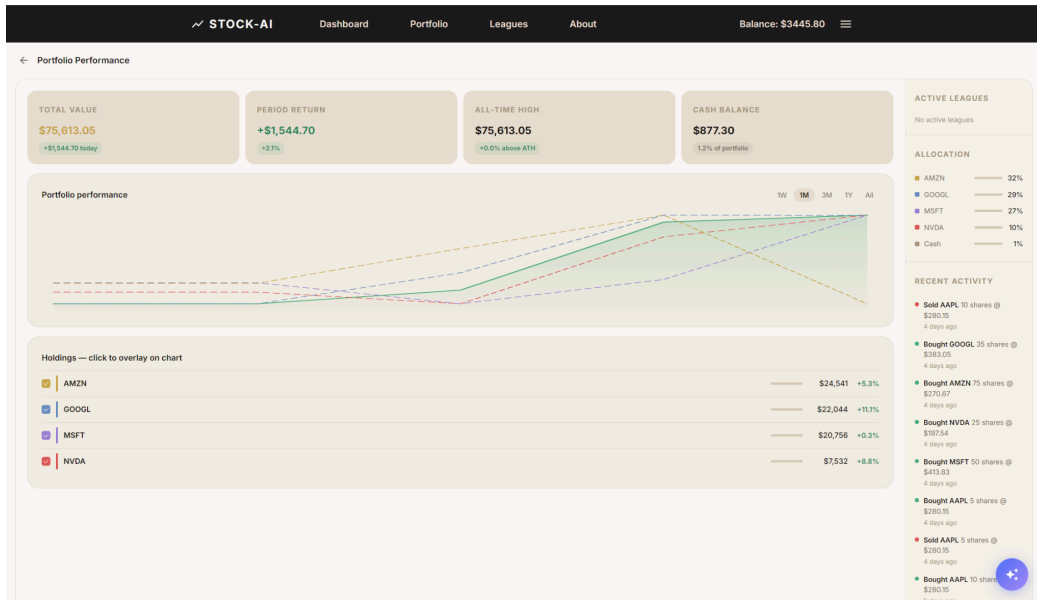


Figure 3: Portfolios performance view. Basic statistics at the top, allocation in the upper right, trade history on the bottom right, and graph view in the middle.

3. Leagues:

Leagues are an optional feature that aims to boost engagement among learners. With it, users can compete against each other by comparing the growth of an entered portfolio. A portfolio is 'snapshot' on entry, and used as a baseline for growth. Comparisons between users are graphed for a visual element. See **Figure 4** below for an example league.

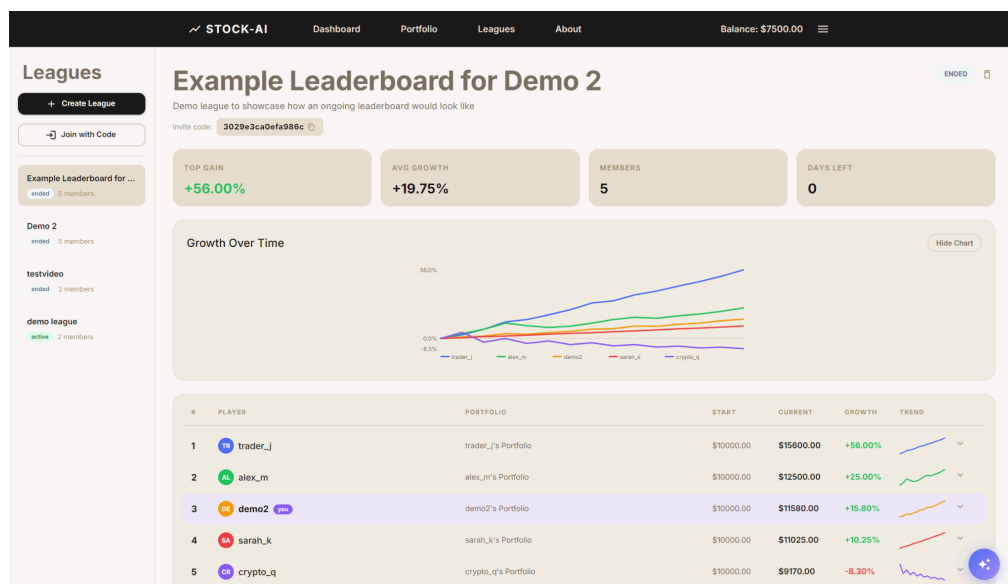


Figure 4: Leagues view, showing an example league with different users' earnings compared.

Design Overview - Tech Stack

Stock-AI follows a fairly standard tech stack for a web application. We are using MariaDB as our database, a FastAPI application to access the database, and a React.js application for our web interface. MariaDB was chosen simply because it was already installed on our server, and it maintains compatibility with standard MySQL. FastAPI was chosen because it offered an easy way to interact with our database and build new endpoints with automatic generation of a Swagger UI documentation page. Similarly, React.js offered an easy way to build out a web UI with different components to be pieced together, which aided the development and implementation of new features. We also utilize other technologies to aid development, such as Podman, Material UI, and Gitlab Runner.

Design Overview - External Resources

For stock data, we are utilizing the yfinance library. It gives Stock-AI both current and historical data to display to the user, and is free to use. It is worth noting that yfinance is not the most constantly updated source of information, due to its nature as a data scraper. This is acceptable for a few reasons; first, Stock-AI is an educational platform, so complete information is more important than down to the second accuracy. Second, Stock-AI is intended to be free, and sourcing more accurate and fully real-time data would incur a non-insignificant cost. Given the breadth of available information through yfinance, it's a worthy tradeoff.

For specific company information such as name, ticker, company logo, description, etc, we are using the FintHub API. It allows us to display a more professional synopsis of a given company, as well as list out news articles that may be relevant for predicting the stock price.

The AI integration in Stock-AI is powered by OpenAI's API. All user prompts/requests are routed through our FastAPI application, which injects context into the prompt to steer the LLM towards what it needs to do - e.g., if a user is asking about AAPL (Apple), the LLM is prompted with "You are a financial advisor specializing in AAPL. The user is asking about AAPL - keep all answers relevant to this stock. Only respond to questions about finance, stocks, and investing. Politely decline any off-topic questions". The result is that the LLM stays on track within the given context.

Implementation - Frontend

Figure 5 pictured below is a diagram of the app's main React components.

AuthContext.tsx wraps the entire app, making authentication available globally. protectedRoute.tsx prevents the user from navigating past the login and register pages if they are not authenticated. A user becomes authenticated by receiving a JWT from the backend after a successful login. The JWT is included in all subsequent requests sent to the backend. **Figure 6** on the next page demonstrates the flow paths for different users.

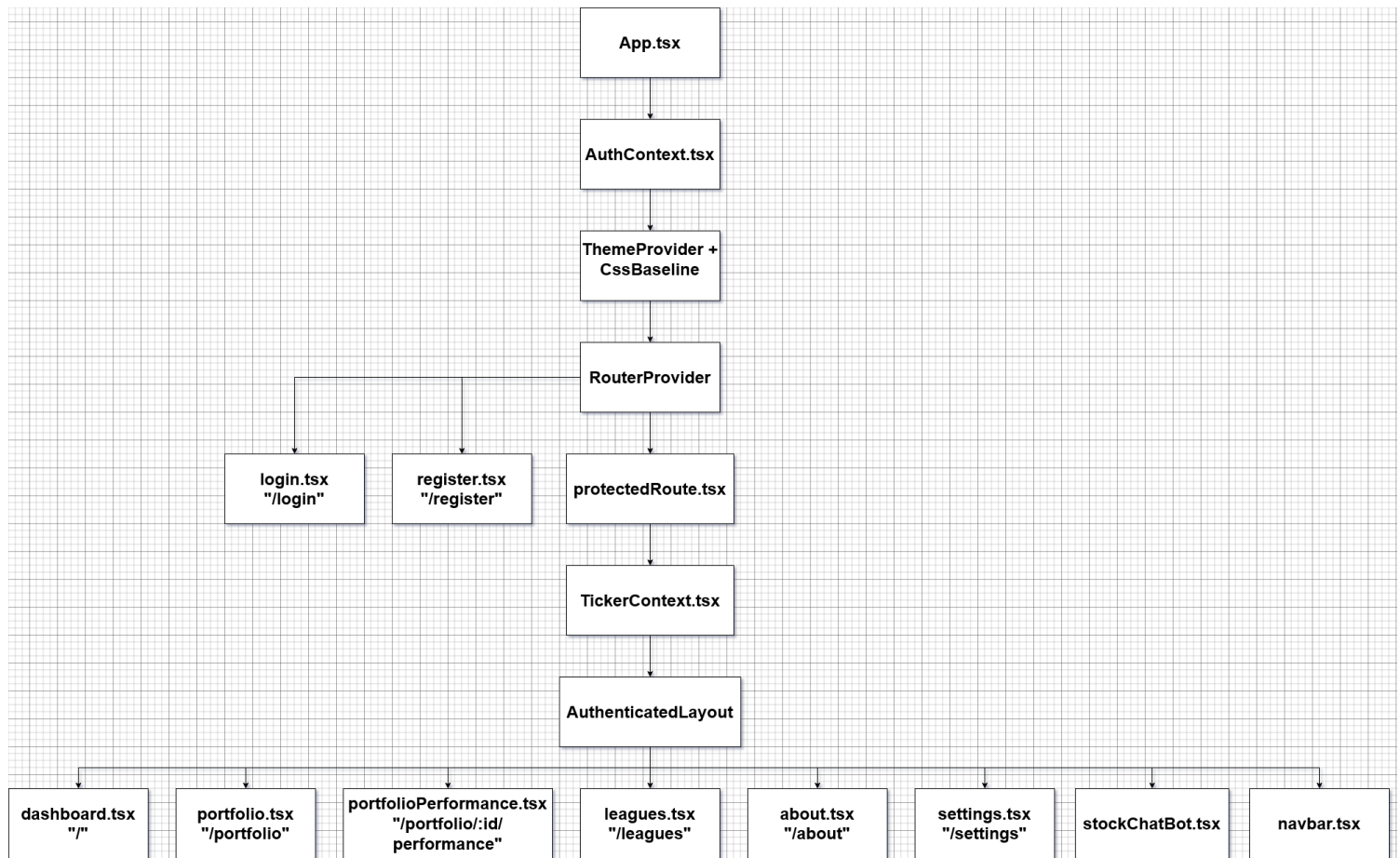


Figure 5: Diagram showcasing the React component structure.

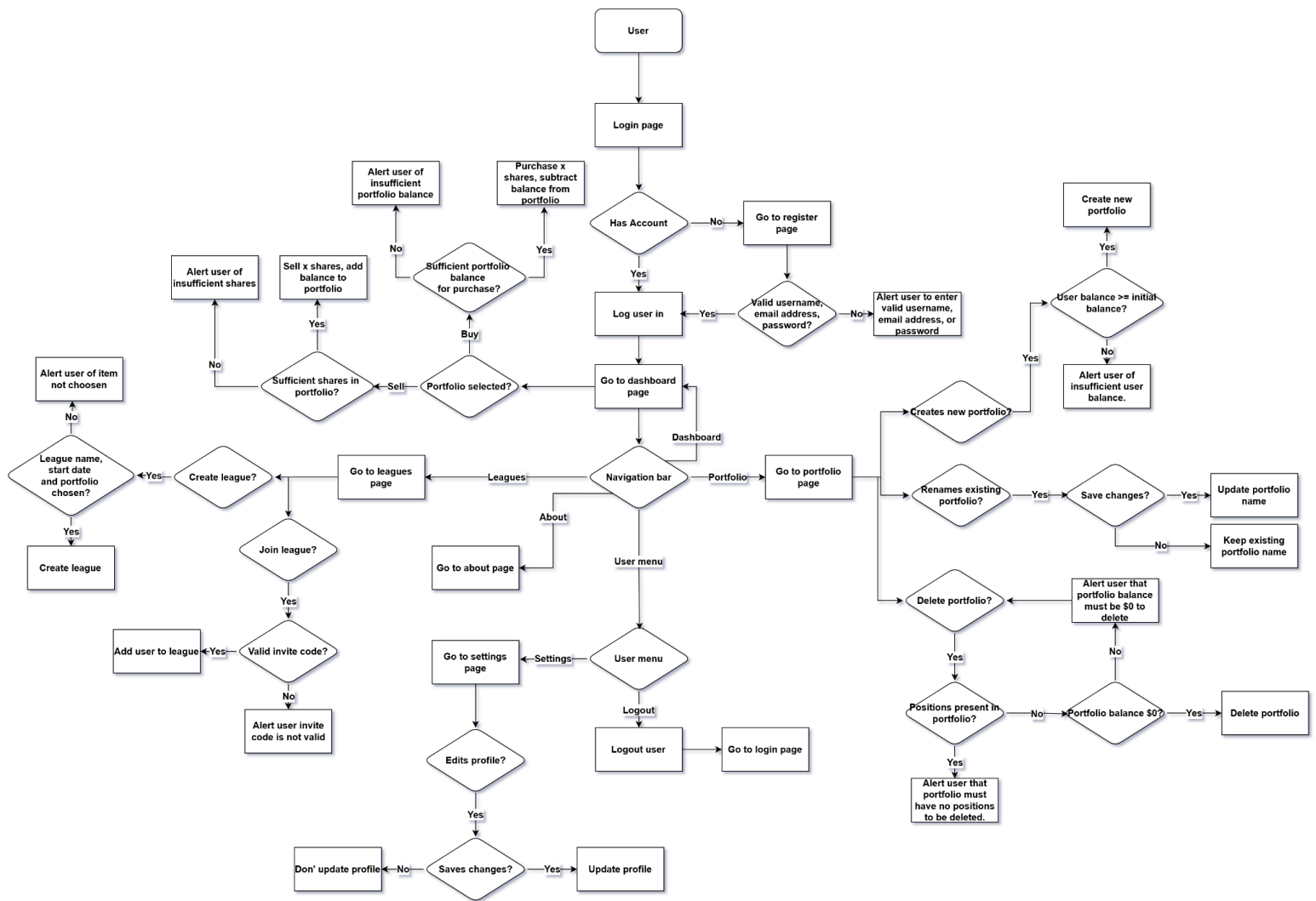


Figure 6: User flow diagram.

Implementation - Backend API

The backend API has many endpoints under multiple categories. For convenience, they are broken down into their respective categories. Category titles serve as the root endpoint for each; i.e., 'get' under 'users' would be '/users/get' if attempting access.

Users

Endpoint	Method	Description
/create	POST	Creates User
/uinfo	GET	Returns basic information about the user.
/login	POST	Log a user in
/protected	GET	Test JWT validity
/update	PATCH	Update user credentials
/delete	DELETE	Soft delete (deactivate) user

Portfolios

Endpoint	Method	Description
/create	POST	Creates new portfolio for user
/rename	PATCH	Renames a portfolio
/{"portfolio_id"}/delete	DELETE	Soft delete (deactivate) portfolio
/listall	GET	Lists all of user's portfolios in JSON
/{"portfolio_id"}/positions	GET	Lists held positions of portfolio in JSON
/cash_in	PUT	Transfers funds from user to portfolio
/cash_out	PUT	Transfers funds from portfolio to user
/positiontransfer	POST	Transfers a full/partial position from one portfolio to another
/{"portfolio_id"}/snapshots	GET	Get portfolio performance snapshots
/{"portfolio_id"}/activity	GET	Lists recent trades by portfolio in JSON

Data

Endpoint	Method	Description
/ {ticker} profile	GET	Read stock profile
/ {ticker} /history	GET	Get stock history
/ {ticker} /news	GET	Get stock news

Trades

Endpoint	Method	Description
/newtrade	POST	Performs a mock trade
/ {portfolio_id} /history	GET	Returns list of trades performed under portfolio in JSON

Chat

Endpoint	Method	Description
/conversation/new	POST	Creates a new conversation
/conversation/rename	PATCH	Rename a conversation
/conversation/latest	GET	Gets users most recent conversation, creating one if none exists
/conversation/ {conversation_id}	GET	Gets chat history for a given conversation
/conversation/send	POST	Save user message and return AI reply

AI

Endpoint	Method	Description
/analyze	POST	Trigger AI analysis of a stock

Leagues

Endpoint	Method	Description
/create	POST	Creates a new league
/mine	GET	Lists all leagues the user has entered
/preview/{invite_code}	GET	Preview a league by invite code
/ {league_id}	GET	Get league details
/ {league_id}	DELETE	Delete given league
/ {league_id}/join	POST	Join a league with a chosen portfolio
/ {league_id}/leaderboard	GET	Get ranked leaderboard for given league
/ {league_id}/snapshots	POST	Trigger a snapshot for all members of given league
/ {league_id}/members/{member_id}/snapshots	GET	Get snapshot time-series for a single member

Internal

Endpoint	Method	Description
/snapshots/daily	POST	Daily portfolio snapshot

Implementation - Database

Figure 7 below demonstrates the database tables, their fields, and the relationships that they have with one another.

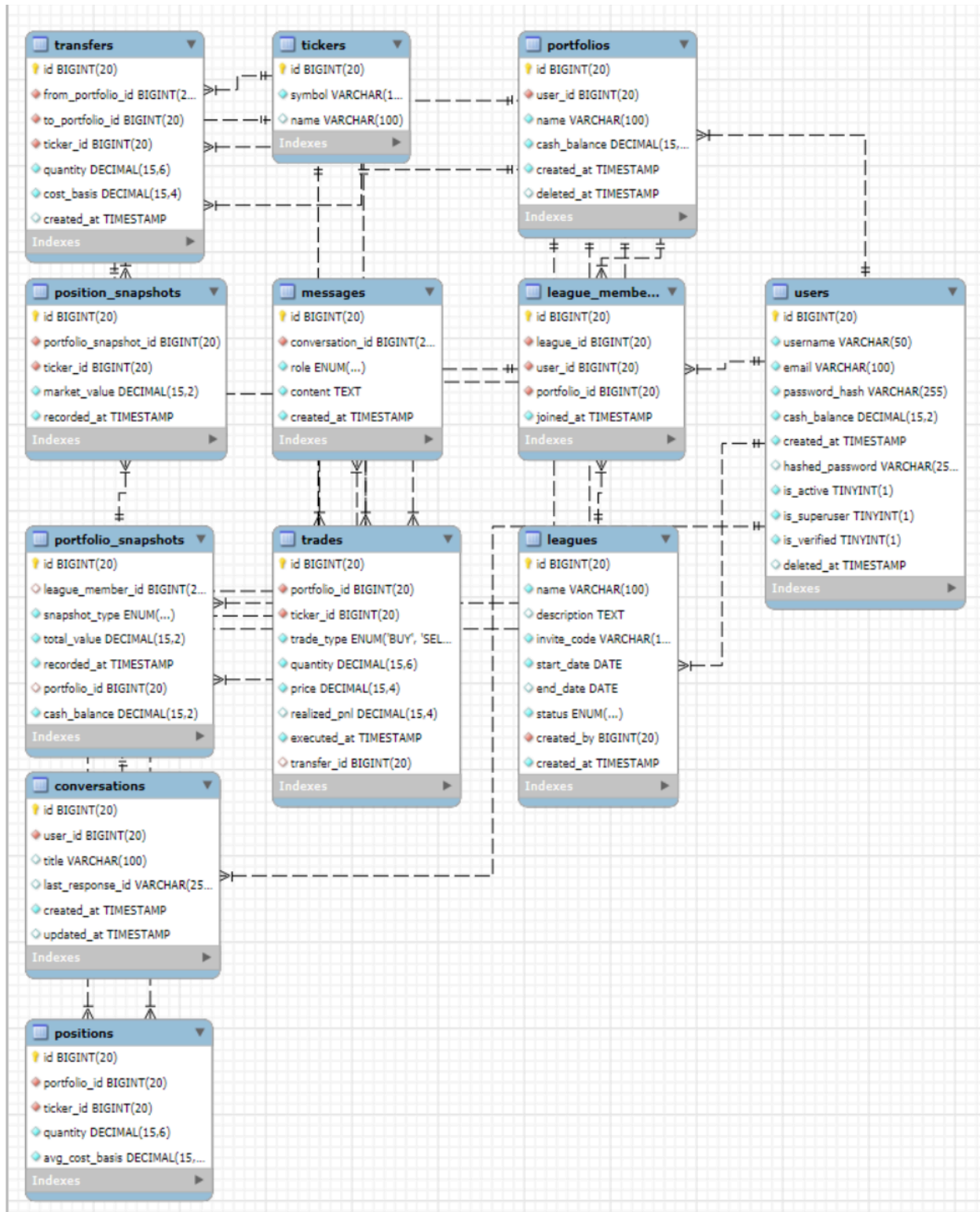


Figure 7: Automatically generated ER diagram via MySQL Workbench.

Work Done

Andrew Butler - Backend development, Client Communication

My first task for the project was to establish basic functionality for the backend - this meant setting up both the MariaDB instance as well as the FastAPI application that would turn into our API. Through the first demo, I had basic user functionality implemented with JWT-based authentication, the beginnings of user portfolios, and a very suboptimal manual deployment using Tmux. The first work for Demo 2 that I did was to learn Docker and Podman to have containerized deployment for the FastAPI application, which enabled us to get CI/CD working so that updates would automatically be pushed to the server. I also flushed out portfolios and trades, enabling mock trading and portfolio management on the backend. I created the initial backend version of conversation history, although it was later modified by Aaron for actual use. My work with demo 3 was lighter; I added the ability for users to update their information and for positions to be transferred between portfolios (sell on donor, buy on destination). I experimented with refactoring our backend to use the more standardized FastAPI Users library for authentication instead of my manually implemented version, but it was incompatible with our existing database relationships, meaning it required a complete rebuild. Throughout the project I also served as the main point of contact with Dr. Phan, scheduling meetings, emailing questions, and communicating progress.

Matthew He - Full Stack development

My contributions to the project spanned both frontend and backend development across all three demos. For Demo 1, I laid out an initial concept and prototype for the stock buying and selling UI, establishing a foundation for how users would interact with trading, though at this stage it was not yet connected to the backend. For Demo 2, I led a redesign of the application's UI to give it a more modern and polished look and feel, helping bring the visual identity of the platform up to the standard the team was aiming for. Also during Demo 2, I implemented the Leagues feature from the ground up on both the frontend and backend, including creating and joining leagues, invite code functionality, portfolio snapshotting on entry to establish a growth baseline, and the leaderboard view that graphs and ranks competitors' portfolio performance over time. For Demo 3, I implemented the Portfolio Performance page, which gives users a detailed view of their portfolio value over time, including a configurable graph with selectable time periods and position overlays, stock allocation breakdown, and a trade history log.

Ethan Roe - Frontend development

My focus for this project was on developing our frontend. My tasks for Demo 1 included creating the first version of our login/signup page and the portfolio page. I worked in tandem with Andrew, who developed the necessary endpoints for those pages in our backend. This also involved creating protected routes for our project, making sure users could not access any pages other than the login and signup pages unless they were logged in. For Demo 2, my main focus was on improving the UI/UX of the portfolios page, since the initial design prioritized functionality over design. I also worked together with Aaron on refactoring our frontend. The refactor mainly focused on breaking up our frontend pages into React components. This made it easier to make changes and add new features to our existing pages moving forward. I also worked on adding the ability to buy and sell stocks within portfolios. For Demo 3, I focused on improving the portfolio's functionality. This included adding the ability to transfer positions between portfolios and the ability to rename and delete portfolios. I also worked on adding a settings menu so users could modify their account details, specifically their username, email, and password.

Aaron Trelstad - Full-Stack Development, AI Integration

My first task for Demo 1 was creating the dashboard, this is the main landing page which enables users to research and analyze different stocks. This involved both frontend and backend work, using the yfinance API for fetching stock related information such as historical prices, recent news, and general company information. For Demo 2, I worked on UI improvements for the dashboard to align it with the theme of the rest of the website and improve user experience. I also started working on how we can integrate AI into our application using the OpenAI API. I created an AI chatbot that took up a static position on the dashboard, it didn't save message history yet. It took in the context of the stock that the user is currently looking at to answer questions related to that specific stock. I also added AI analytics, which gave AI generated signals like BUY/SELL, Risk, Momentum and a Summary. This was designed to help the user get a well rounded perspective of the stock. For Demo 3, I worked on enhancing the AI features, I changed the AI Chatbot to be opened by a button that is present on every screen in the application, this makes the feature more accessible to users. I also added message history which allows a more personalized experience for users. Another improvement was adding explainability to the AI generated analytics, like how/why did it give the BUY/SELL signal or evaluate the current Risk, this helps to give the user more context.

Results

1. Successes:

Stock-AI works well as a free learning platform; it was built with minimal expenses, only requiring sufficient host infrastructure and OpenAI API access. Stock value lookup and historical data is sufficient for an educational focus, with yfinance giving a wide range of available options. Our user portfolios allow flexible organization of positions, as well as useful performance tracking. Leagues also provided a unique take on incentivising learning by provoking a competitive spirit among users. AI analysis serves as a decent indicator for company direction, and the AI chat feature allows persistent conversations that are held on track by our prompt engineering.

2. Could be Improved:

The Stock-AI API, while functional, could use a slight redesign of endpoints to make it more consistent with other applications. Most notably, the inclusion of FastAPI Users or a similar standardized structure. Additionally, the use of the yfinance library as the primary source of stock data introduces inaccuracies, as discussed above. While sufficient for this project at this stage, a more professional deployment would get stock data directly and in real time to gain as much accuracy as possible. Similarly, the non-deterministic nature of AI models could prove confusing for new learners.

Stock Trading Platform With AI-Integration:

Suggested Features

Team UG_HF_03

Andrew Butler, Matthew He, Ethan Roe, & Aaron Trelstad

Basic Website Features

- User accounts + Login
 - Login, signup
 - Password hashing
 - Session expiration & logout handling
- 'Guest' mode for users who have not created an account
- About page w/ team & project information

Market Inspection

- View overall market status (S&P, Dow, Nasdaq)
 - Trend detection using moving averages
- For any given U.S. Stock
 - Current price
 - Daily high/low
 - 5-day, 1-month charts
- Watchlist
 - Pin certain stocks for quick status viewing

Stock Trading

- Mock purchases
 - Save share price, amount purchased to user portfolio
 - Users start with x amount of 'money' to spend
- Transaction history
- User cash balance tracking

Portfolio Dashboard

- Total portfolio value
- Unrealized gain/loss
- Per-stock breakdown

AI Integration

- AI Chat feature; allow users to ask questions
 - E.g., "should I buy Nvidia?"
 - E.g., "Explain what a call option is."
- Trade Recommendations (AI opinion on trade over specific stock)
 - Bull/Bear case
 - Risk level
 - AI suggested action: buy, hold, or sell

Additional Feature Ideas

- User-configurable color organization of portfolios
- Stock monitoring w/ push notifications to users (price high, low)
- Account updates: password/email change
- Notification center on website so users can see any important updates (such as price changes) that have occurred while logged in / logged out.
- Custom user groups w/ leaderboard for who has earned most
- Graph portfolio value history

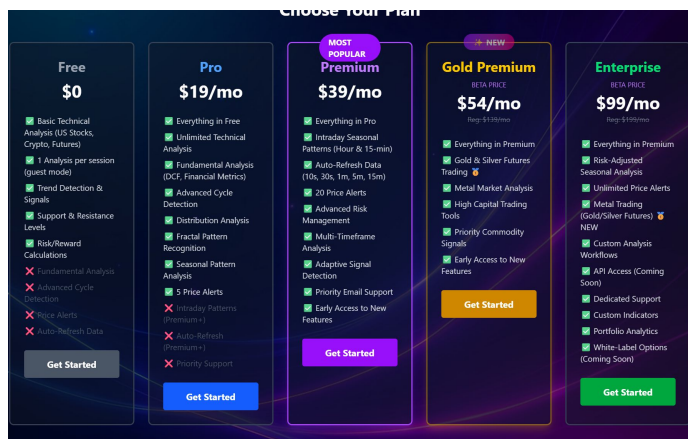
Stock-AI Demo #1

Team UG_HF_03

Andrew Butler, Matthew He, Ethan Roe, Aaron Trelstad

Problem

- Stock trading is complicated
- Learning the market is difficult
- Platforms with learning focus exist, but are inaccessible
 - Use proprietary LLMs
 - No free service, often very expensive



Solution

- Use AI to provide intuitive & interactive help with trading
- Make our AI-integrated platform free & accessible

Two-pronged development

- Need to build AI Agent capable of helping users
- Need stock trading platform nuanced enough to integrate & evaluate AI

Team task breakdown

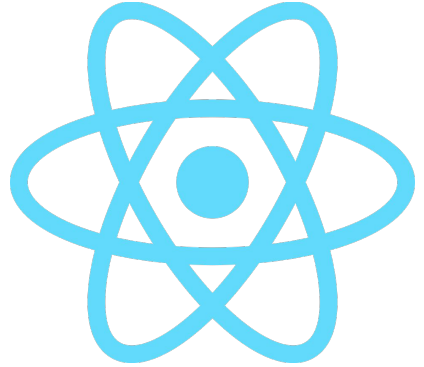
- Graduate Team
 - Building & refining open-source model for stock trading
 - Produce thoughts/recommendations on a given trade
 - Chat with model
- Undergraduate Team (us)
 - Create web platform for mock stock trading
 - User accounts with portfolios, ability to inspect market & make mock trades
 - Integrate features of LLM into user experience

Development problems

- Real-time data access
 - Solution: yfinance websocket, free & unlimited use
- Secure user accounts + user actions
 - Hashed password storage + JSON Web Tokens (JWTs)
- Backend downtime & Deployment issues
 - Prevented frontend-backend connection
 - Containerization of backend to streamline deployment

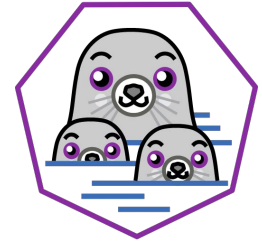
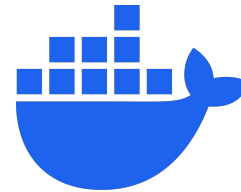
Technologies in use - Frontend

- **React.js**
 - Web framework
 - Templates & automatic HTML generation
- **Lightweight Charts**
 - Visualizing stock data
- **Material UI**
 - Component Library



Technologies in use - Backend

- FastAPI Python framework
 - Easy creation of endpoints
 - Automatic Jagger documentation
- MariaDB
 - MySQL database
- Docker / Podman
 - Local & server containerization, respectively
- Insomnia / Postman
 - Testing endpoints



Project Status

- User Authentication
- Protected Routes

STOCK-AI API openapi.com GET OAuth2.0

This is the API for Stock-AI, an AI-integrated stock trading platform. Currently a work in progress, endpoints & functionality are subject to change.

[Authorize](#)

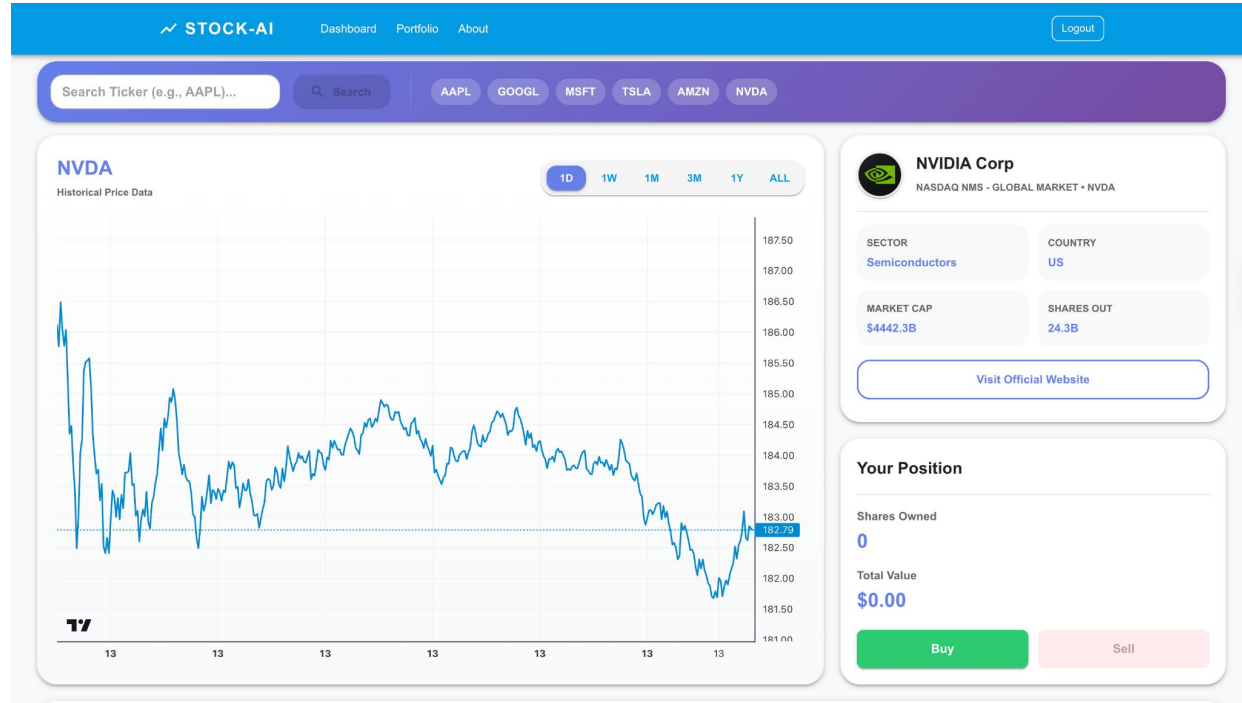
users		^
POST	<code>/users/create</code>	User account creation / signup
POST	<code>/users/login</code>	User account login
GET	<code>/users/protected</code>	Test JWT validity
portfolios		^
POST	<code>/portfolios/create</code>	Create a new portfolio for user
GET	<code>/portfolios/listall</code>	Grab all portfolios for user
data		^
GET	<code>/data/{ticker}/profile</code>	Read Stock Profile
GET	<code>/data/{ticker}/history</code>	Get Stock History
GET	<code>/data/{ticker}/news</code>	Get Stock News Endpoint
default		^
GET	<code>/</code>	root request
GET	<code>/num_users</code>	Grab number of users listed in database

Login Page

[Login](#)[Register](#)

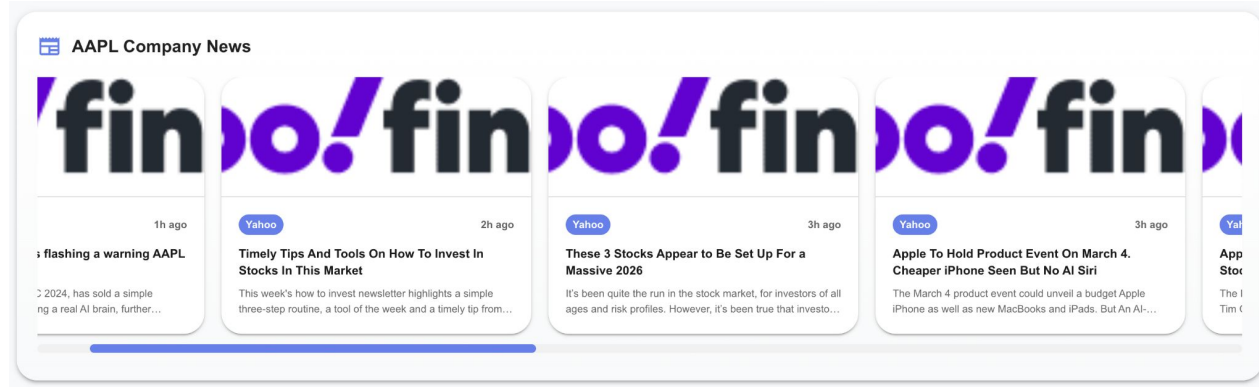
Project Status

- Stock dashboard
 - Ticker lookup
 - Historical prices
 - Company profile
 - Buy/Sell positions



Project Status

- Stock dashboard
 - Company News



Project Status

- Stock Portfolio

My Portfolios

[Create](#)

Test (Created: 2/16/2026)

Project Status

- About Page

A Tool to Help Anyone Practice Investing Stocks

Requested by Dr Hung D Phan, this website is a senior design project developed to help lorem ipsum stuff and stuff.



All
Stocks Supported

0
AI Features

Quick Demo

What's next?

- Expanding on portfolio functionality
 - Allow users to buy/sell
- AI Integrations
 - Give users recommendations

Challenges

- AI Integration Ambiguity
- Error handling
- Communication

Timeline for remaining features

- Demo 2
 - Features
 - Watchlist, Cash Balance, Transaction History, More Detailed Portfolio
 - This Week
 - Plan tickets and responsibilities
- Demo 3
 - Features
 - AI Chat features and suggestions
 - Responsive UI that deals with AI Suggestions

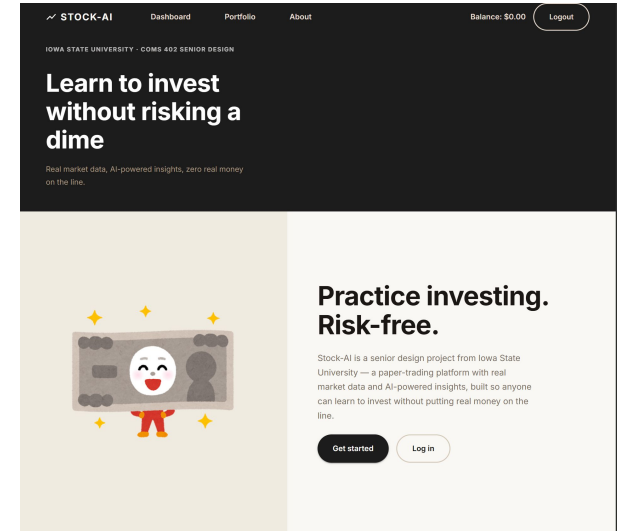
Stock-AI Demo #2

Team UG_HF_03

Andrew Butler, Matthew He, Ethan Roe, Aaron Trelstad

What is Stock-AI?

- Learning stock trading is difficult
- Trading & trading platforms are expensive
- Create Free learning platform
 - Mock trades, so no risk
 - AI assistance (ask questions, provide recommendations)



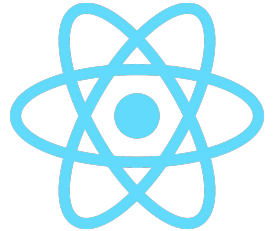
Technologies in use

Backend

- MariaDB
- FastAPI Python framework
- Podman for containerized deployment

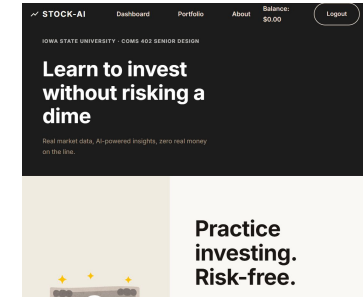
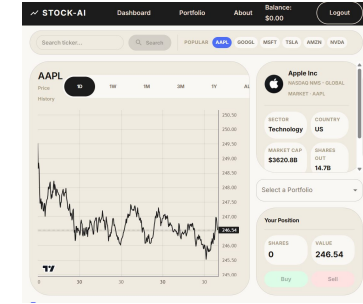
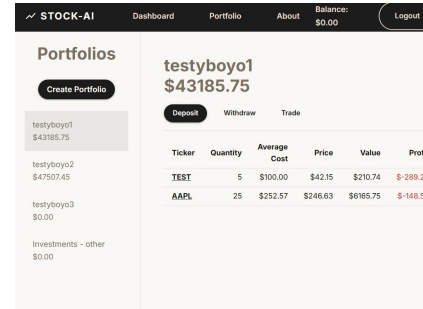
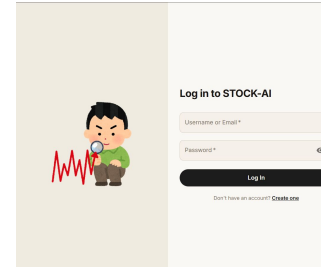
Frontend

- React.js web framework
- Material UI component library



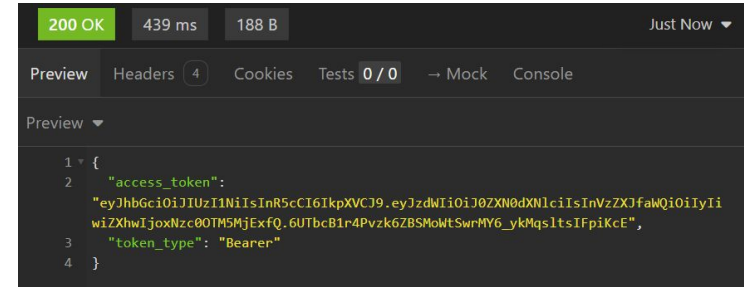
Stock-AI Functionality

- Typical website design
 - Login, register page
 - Dashboard page
 - Portfolios page
 - Leagues page
 - About page



Implementation - Login & Security

- Password hashing
- Login generates Json Web Token (JWT)
 - Gives token
 - Contains user ID, username
 - Validates future action
 - Expires after 60 minutes
- Protected routes
 - Anything that is user specific requires valid JWT
 - Endpoints check JWT validity
 - Endpoints check user ownership of resource (portfolio, conversation, etc)
- Suboptimal: Implemented manually



The screenshot shows a REST client interface with a status bar at the top indicating a 200 OK response, a response time of 439 ms, and a body size of 188 B. The 'Preview' tab is active, displaying a JSON object with an 'access_token' and a 'token_type' of 'Bearer'. The token is a long alphanumeric string.

```
1 {  
2   "access_token":  
   "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiI0ZDZlN0dXNlciIsInVzZXI6ImFhQWQiOiIyIiwiaXNjaXhwIjoxNzc0OTM5MjExfQ.6UTbcB1r4Pvzk6ZBSMoWtSwrMY6_ykMqs1tsIFpiKcE",  
3   "token_type": "Bearer"  
4 }
```

Implementation - Trading

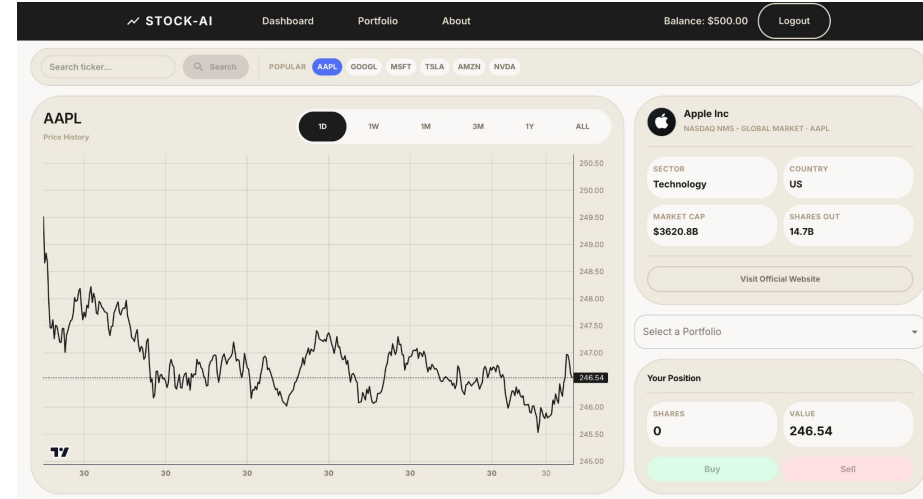
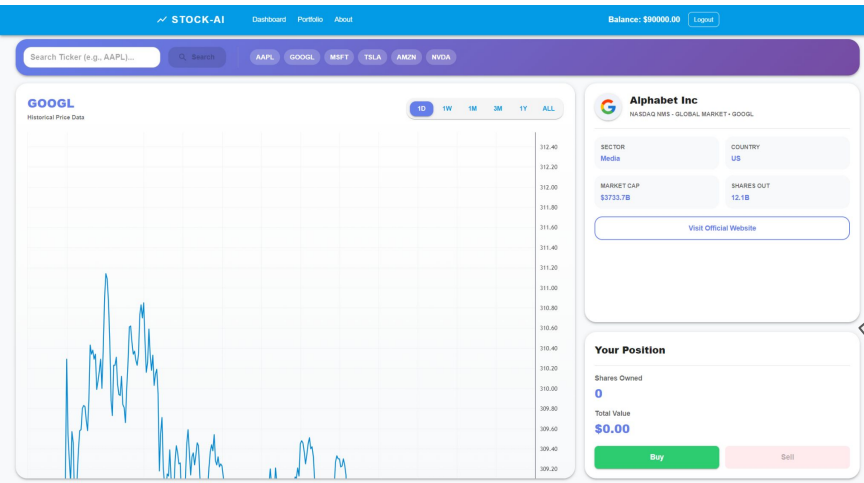
- Users can have multiple portfolios
- User & Portfolio cash balances are handled separately
 - User can transfer money in & out of a given portfolio
- What data we store:
 - User & portfolio cash balance
 - Held positions
 - Trade history (eventually viewable)
 - Average cost basis for position
 - Ticker symbol (e.g., AAPL)

	id	symbol	name
▶	1	TEST	TestCo
	3	AAPL	Apple Inc
	4	GOOGL	GOOGL
	5	MSFT	Microsoft Corp
	9	NVDA	NVIDIA Corp
	10	TSLA	Tesla Inc
	11	AMZN	Amazon.com Inc
•	NULL	NULL	NULL

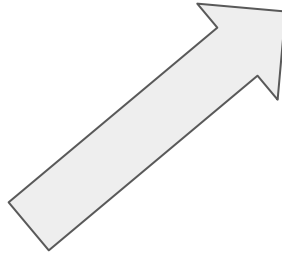
	id	portfolio_id	ticker_id	quantity	avg_cost_basis
▶	6	1	1	5.000000	100.0000
	7	1	3	25.000000	252.5700
	8	5	4	2.000000	0.0000
	12	6	4	1.000000	274.2300
	13	2	9	15.000000	166.1700
	14	4	4	10.000000	274.1700
	15	4	10	20.000000	357.1700
	16	4	9	10.000000	166.2500
	17	17	3	1.000000	246.5400
	19	29	3	5.000000	246.5400
	20	30	5	2.000000	358.7600
	21	29	4	2.000000	273.5100
	22	27	3	5.000000	246.5400
	23	27	4	3.000000	273.5100
	24	27	9	5.000000	165.0600
	25	6	11	3.000000	200.9000
	26	32	3	4.000000	246.5400
•	NULL	NULL	NULL	NULL	NULL

	id	portfolio_id	ticker_id	trade_type	quantity	price	realized_pnl	executed_at
▶	1	4	1	BUY	10.500000	250.0000	NULL	2026-02-24 21:13:02
	2	4	1	BUY	5.000000	250.0000	NULL	2026-02-24 21:24:03
	3	4	1	BUY	5.000000	150.0000	NULL	2026-02-24 21:24:14
	4	4	1	BUY	0.500000	0.0000	NULL	2026-02-24 21:24:39
	5	4	1	BUY	4.000000	-10.0000	NULL	2026-02-24 21:24:54
	6	4	1	BUY	5.000000	-10.0000	NULL	2026-02-24 21:27:11
	7	4	1	BUY	5.000000	0.0000	NULL	2026-02-24 21:34:27
	8	4	1	BUY	0.000000	0.0000	NULL	2026-02-24 21:34:33
	9	4	1	BUY	0.000000	0.0000	NULL	2026-02-24 21:34:36
	10	4	1	BUY	0.000000	0.0000	NULL	2026-02-24 21:34:44
	11	4	1	BUY	100.000000	250.0000	NULL	2026-02-24 21:35:21
	12	4	1	BUY	75.000000	250.0000	NULL	2026-02-24 21:36:29

Dashboard: UI Updates



Dashboard: UI Updates



 AAPL News

1

Yahoo

Just now

Western Digital, Photronics, Nova, FormFactor, and Amkor Stocks Trade Down, What You Need To Know

>

2

Yahoo

Just now

Amtech, Magnachip, Monolithic Power Systems, Allegro MicroSystems, and Kulicke and Soffa Shares Are Falling, What You Need To Know

>

3

Yahoo

Just now

Himax, Teradyne, Entegris, IPG Photonics, and Universal Display Shares Plummet, What You Need To Know

>

4

Yahoo

Just now

Seagate, Applied Materials, KLA Corporation, Lam Research, and MACOM Stocks Trade Down, What You Need To Know

>

5

Yahoo

1h ago

1 Brilliant Growth Stock to Buy Before It Joins Nvidia, Apple, and Alphabet in the \$3 Trillion Club

>

 AAPL Company News



1h ago

flashing a warning AAPL

> 2024, has sold a simple ng a real AI brain, further...



2h ago

Timely Tips And Tools On How To Invest In Stocks In This Market

This week's how to invest newsletter highlights a simple three-step routine, a tool of the week and a timely tip from...



3h ago

These 3 Stocks Appear to Be Set Up For a Massive 2026

It's been quite the run in the stock market, for investors of all ages and risk profiles. However, it's been true that investo...



3h ago

Apple To Hold Product Event On March 4. Cheaper iPhone Seen But No AI Siri

The March 4 product event could unveil a budget Apple iPhone as well as new MacBooks and iPads. But An AI...



App Stor

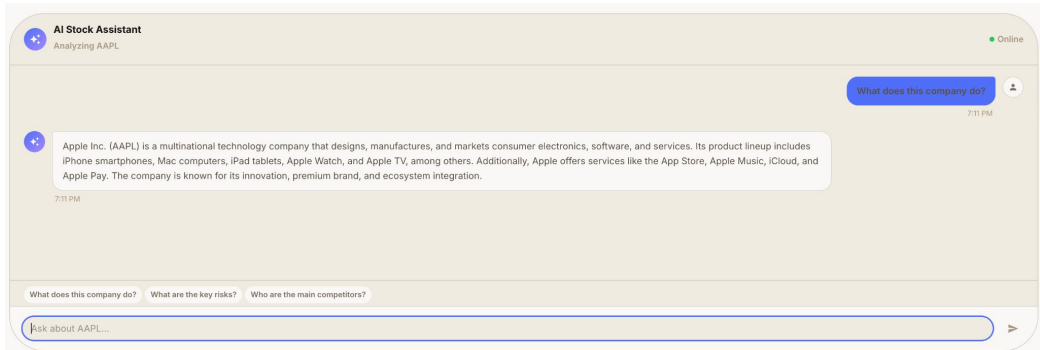
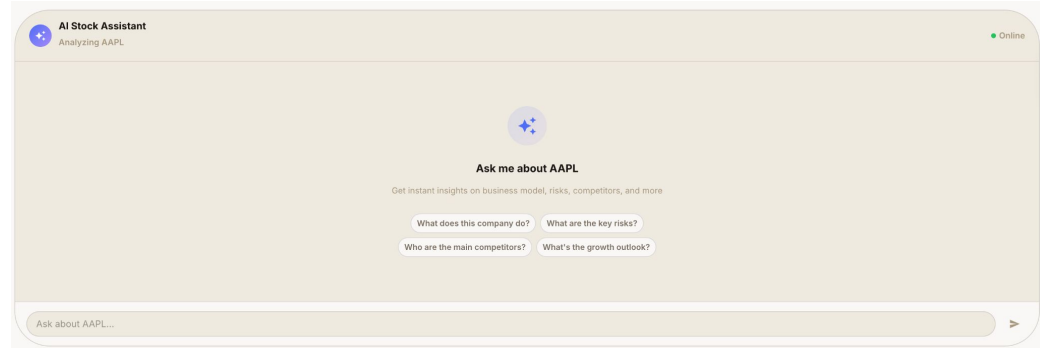
The I

Tim C

IOWA STATE
UNIVERSITY 8/22

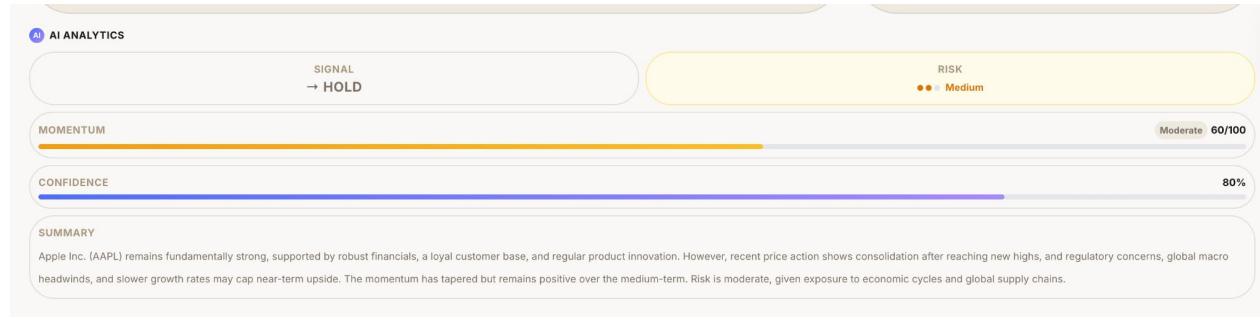
AI Chatbot

- OpenAI Conversation API

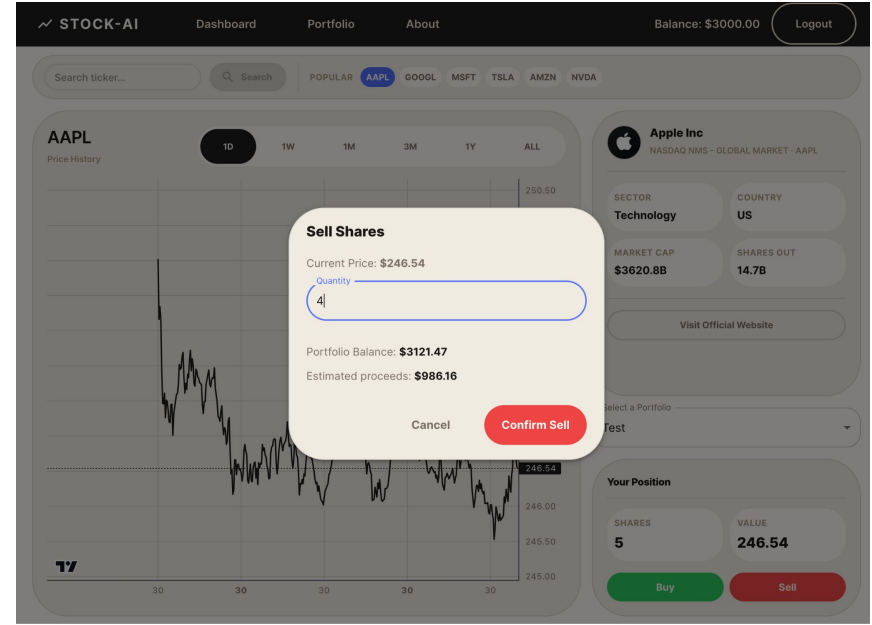
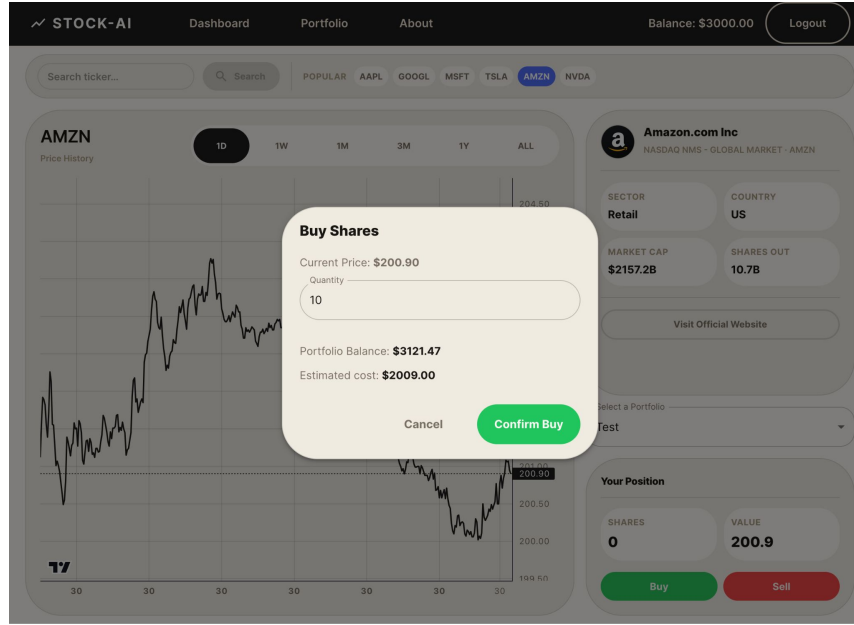


AI Analytics

- OpenAI Agents API

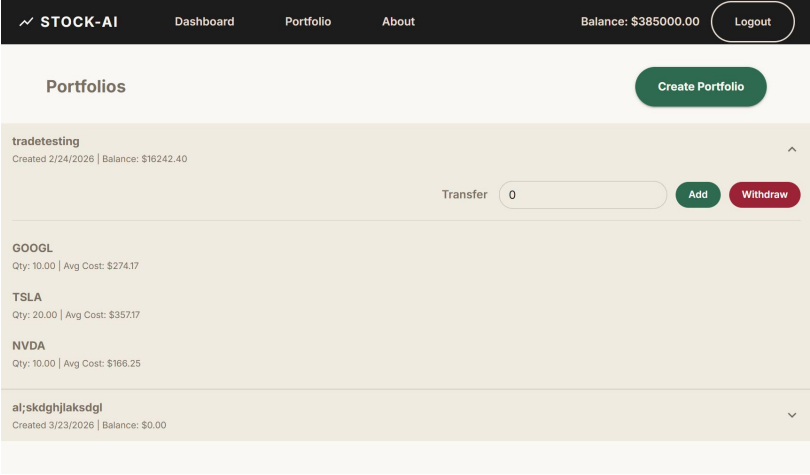


Mock Trading

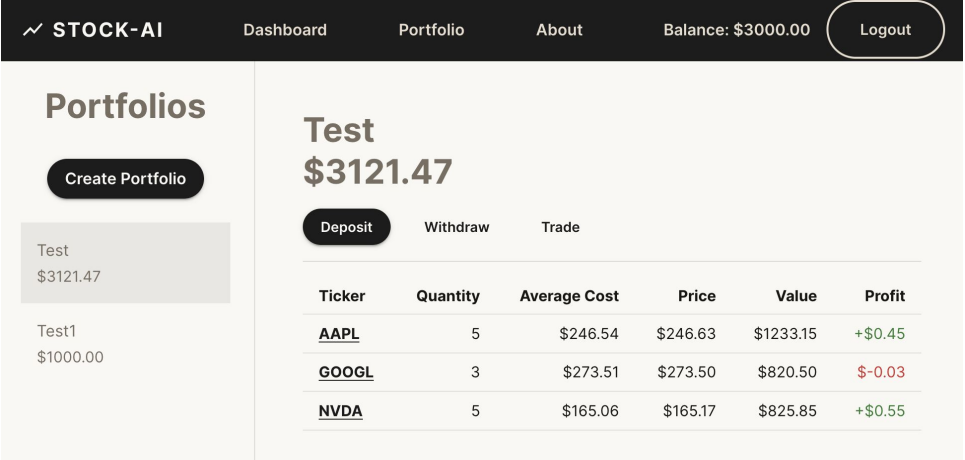


Portfolio UI/UX Improvements

Before:

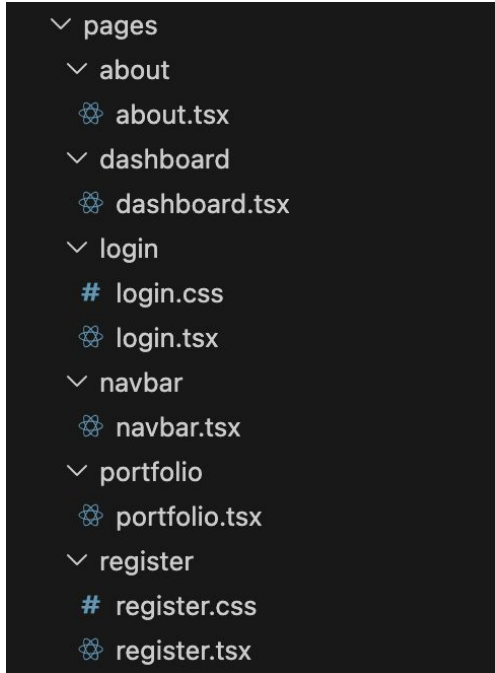


After:

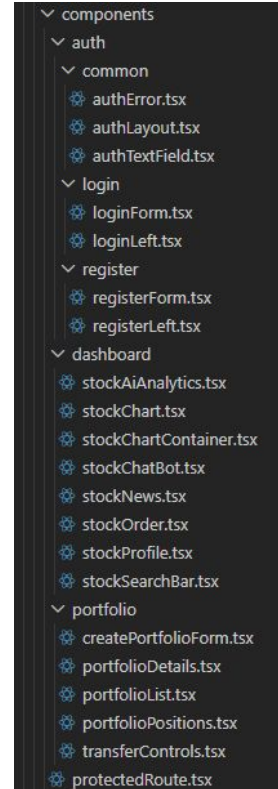


Frontend Code Refactor

Before:

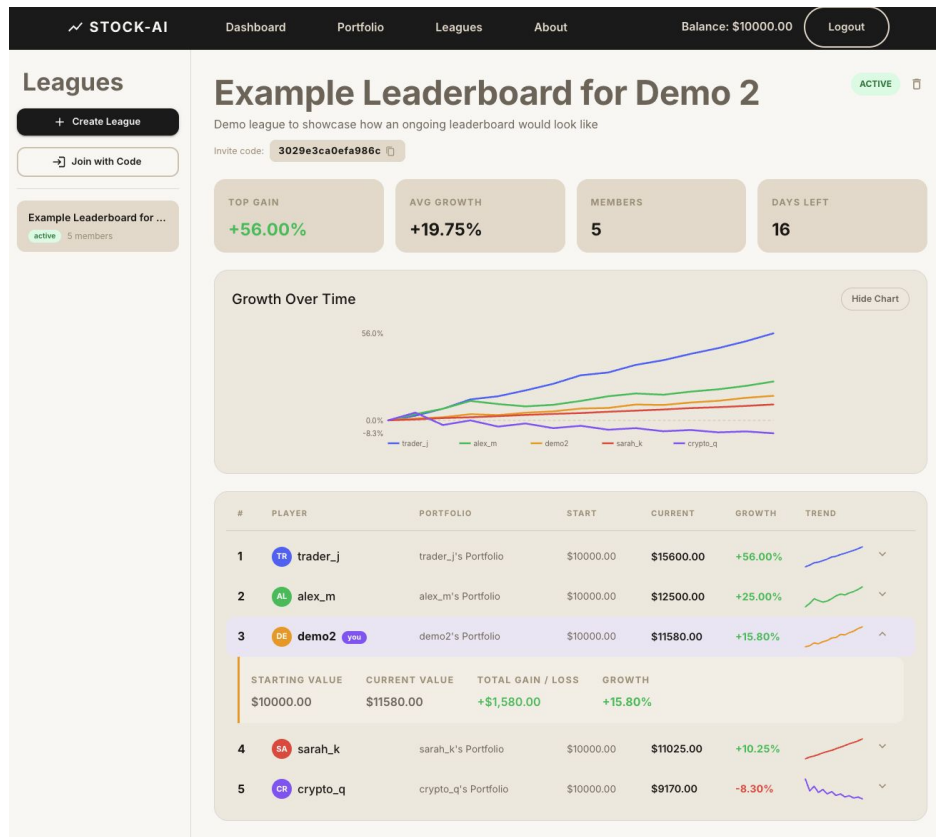


After:

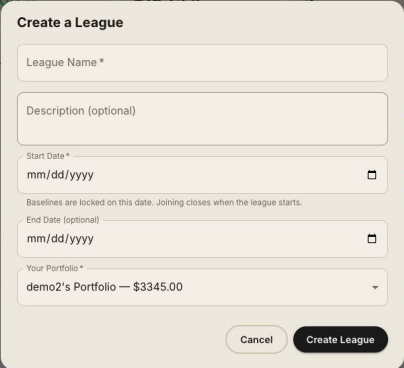


Competitive Leaderboards

- Engaging Stats
- Growth Chart
- Fantasy Sports Inspired Table
 - Sparklines
 - Expandable Rows
- Status Tags



Creating/Joining/Deleting Leaderboards



Create a League

League Name *

Description (optional)

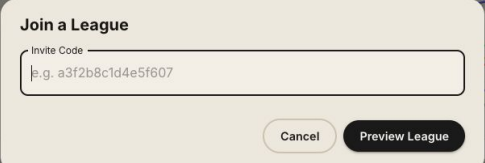
Start Date *
mm/dd/yyyy

Baselines are locked on this date. Joining closes when the league starts.

End Date (optional)
mm/dd/yyyy

Your Portfolio *
demo2's Portfolio — \$3345.00

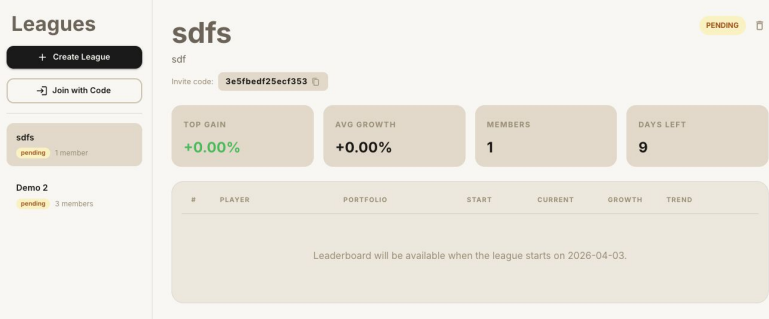
Cancel Create League



Join a League

Invite Code
e.g. a3f2b8c1d4e5f607

Cancel Preview League



Leagues

+ Create League

- Join With Code

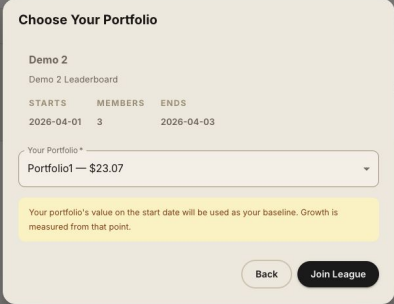
sdfs PENDING

Invite code: 3e5fbedf25ecf353

TOP GAIN +0.00% AVG GROWTH +0.00% MEMBERS 1 DAYS LEFT 9

Demo 2 PENDING 3 members

#	PLAYER	PORTFOLIO	START	CURRENT	GROWTH	TREND
Leaderboard will be available when the league starts on 2026-04-03.						



Choose Your Portfolio

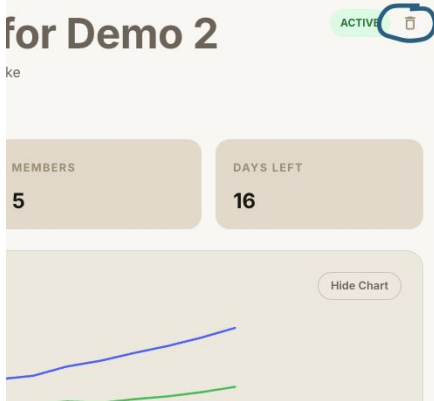
Demo 2
Demo 2 Leaderboard

STARTS	MEMBERS	ENDS
2026-04-01	3	2026-04-03

Your Portfolio *
Portfolio1 — \$23.07

Your portfolio's value on the start date will be used as your baseline. Growth is measured from that point.

Back Join League



for Demo 2 ACTIVE

MEMBERS 5 DAYS LEFT 16

Hide Chart

Competitive Leaderboard Architecture

- New Tables
 - League
 - League Member
 - Portfolio Snapshot
- Routes

leagues			^
POST	/leagues/create	Create a new league	🔒 ▼
GET	/leagues/mine	List leagues the current user belongs to	🔒 ▼
GET	/leagues/preview/{invite_code}	Preview a league by invite code (no auth required)	▼
GET	/leagues/{league_id}	Get league detail	🔒 ▼
DELETE	/leagues/{league_id}	Delete a league	🔒 ▼
POST	/leagues/{league_id}/join	Join a league with a chosen portfolio	🔒 ▼
GET	/leagues/{league_id}/leaderboard	Get ranked leaderboard for a league	🔒 ▼
POST	/leagues/{league_id}/snapshots	Trigger a snapshot for all members of a league	🔒 ▼
GET	/leagues/{league_id}/members/{member_id}/snapshots	Get snapshot time-series for a single member (race chart data)	🔒 ▼

UI redesign - Login/Register

Login

Login Register



Log in to STOCK-AI

Log In

Don't have an account? [Create one](#)

Register

Register Back to Login

Create your account

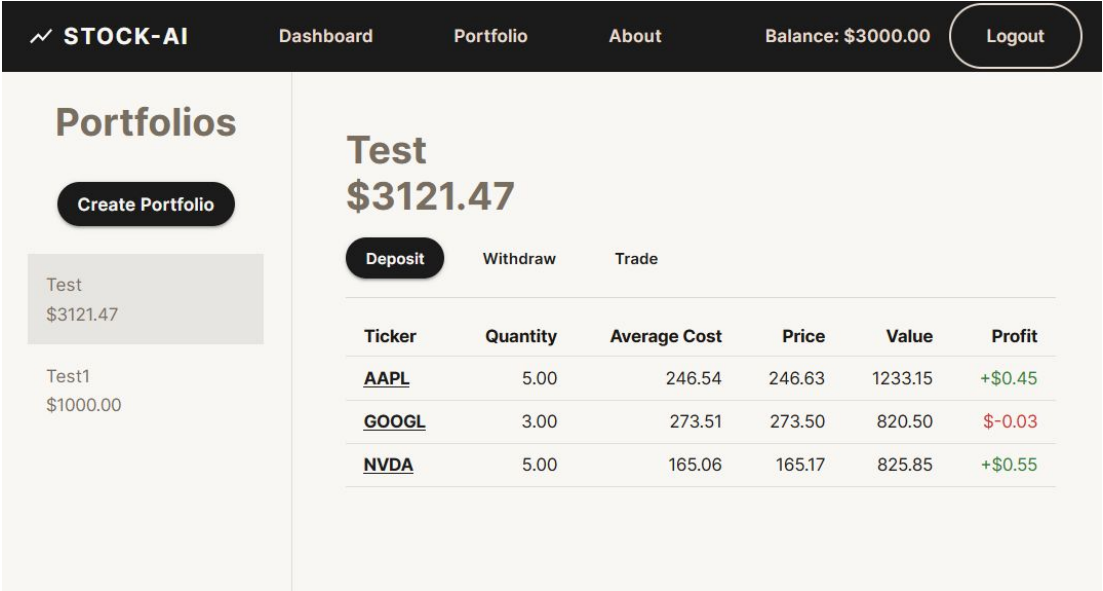
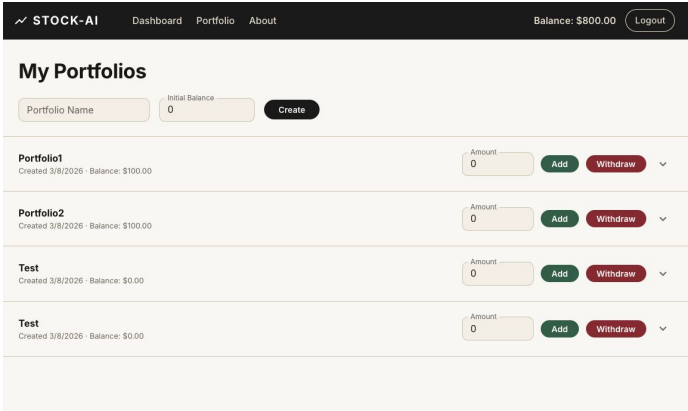
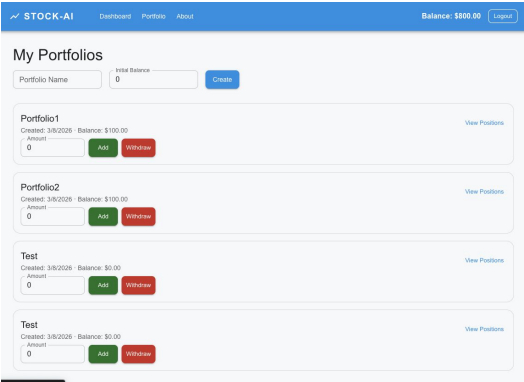
We'll need your username, email address, and a unique password. You'll use this login to access STOCK-AI next time.

Already have an account? [Log In](#)

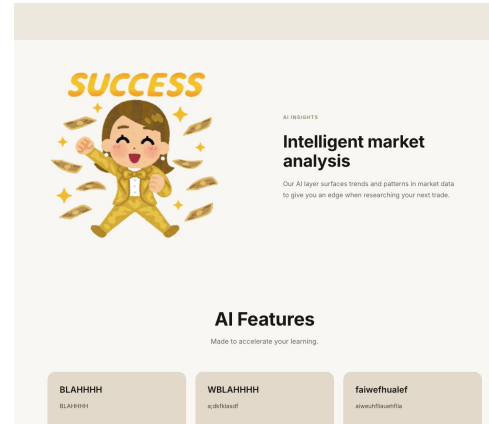
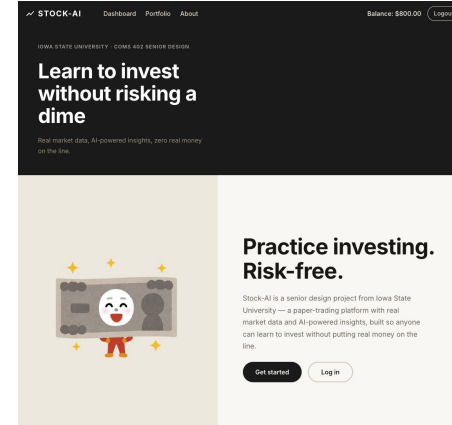
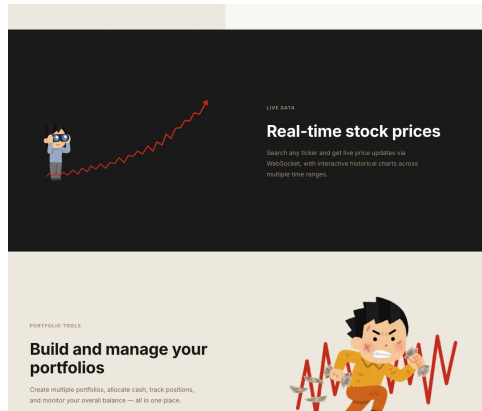
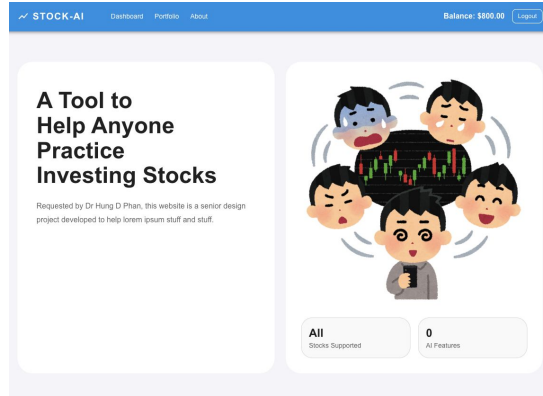
Continue



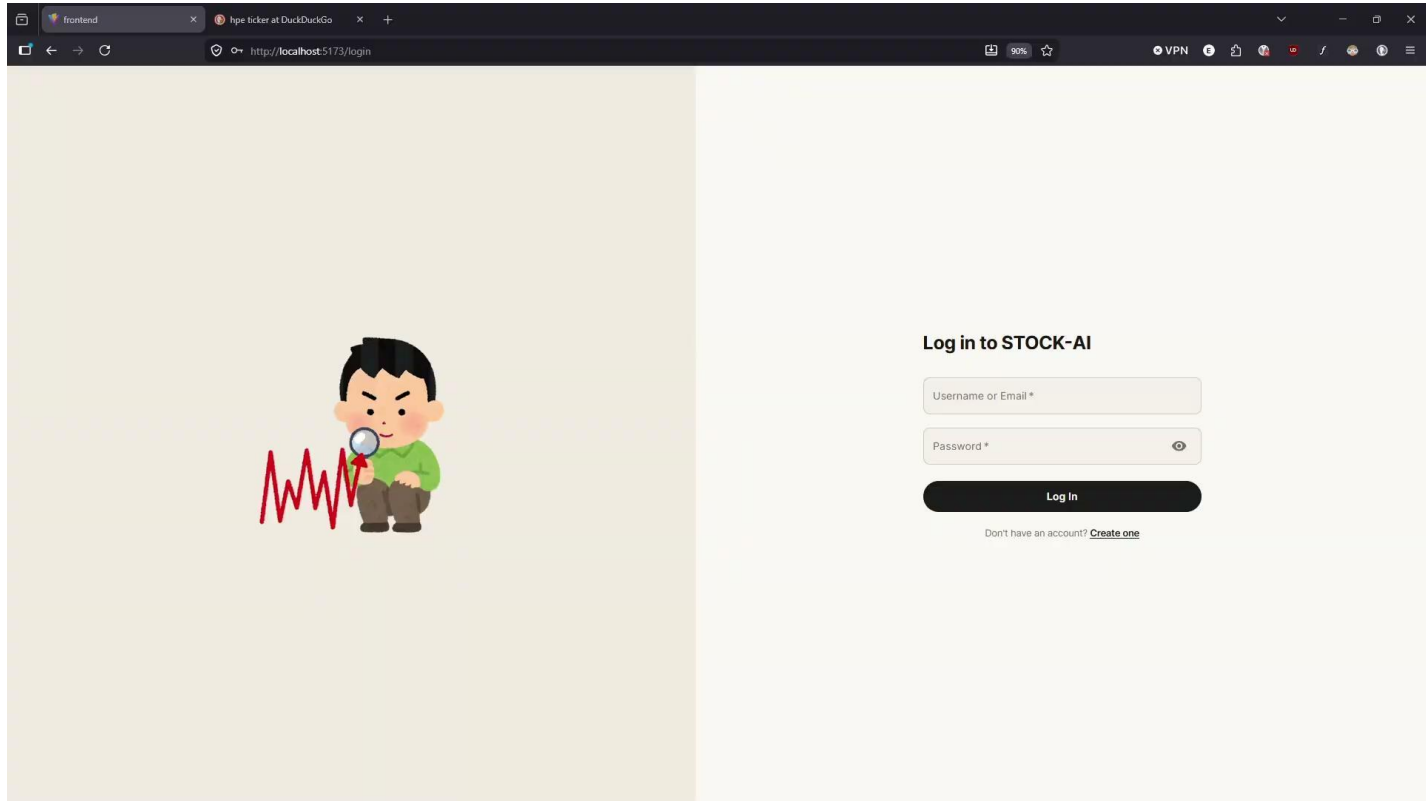
UI redesign - Portfolios



UI redesign - About Page



Demo



[Youtube upload if embed doesn't work](#)

Our goals moving forward

- Security refactor: FastAPI Users library
 - 'Forgot Password' pathway
 - Encrypt portfolio, trade data?
- Custom Watchlists
- Trade & chat history viewing
 - Endpoints already exist; just need UI
- AI model selection / graduate group model
- Further improvements to portfolio
 - Transferring trades between portfolios
 - Ability to delete or rename portfolios

Questions?

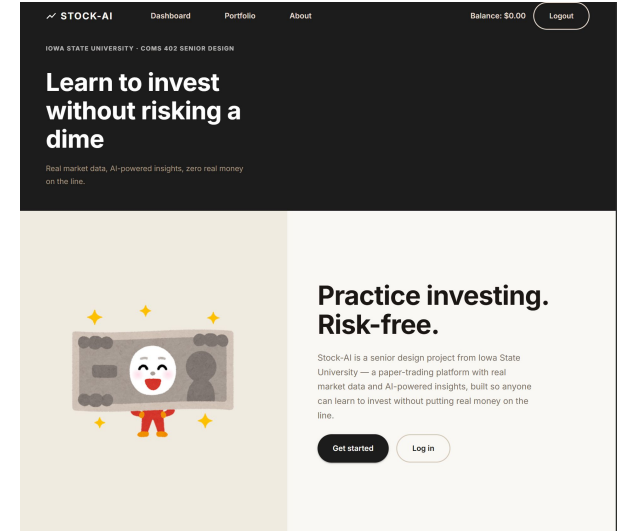
Stock-AI Demo #3

UG_HF_03

Andrew Butler, Matthew He, Ethan Roe, Aaron Trelstad

What is Stock-AI?

- Learning stock trading is difficult
- Trading & trading platforms are expensive
- Create Free learning platform
 - Mock trades, eliminating financial risk
 - AI assistance (ask questions, provide recommendations)



Technologies in use

Backend

- MariaDB
- FastAPI Python framework
- Podman for containerized deployment

Frontend

- React.js web framework
- Material UI component library

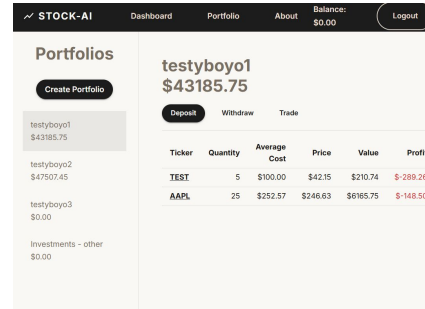
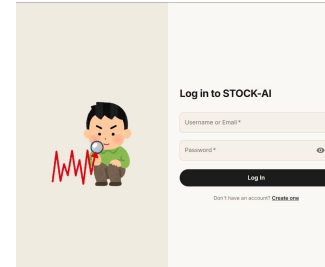
External Resources

- OpenAI API
- yfinance

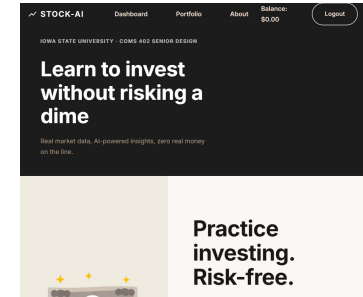
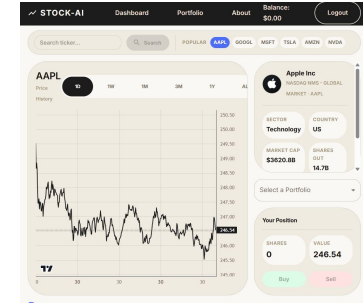


Stock-AI Features

- Secure login + authentication
- User Portfolios w/ value tracking
- Stock Dashboard
- AI Chat assistant
- AI Trading recommendations
- Competitive Leagues

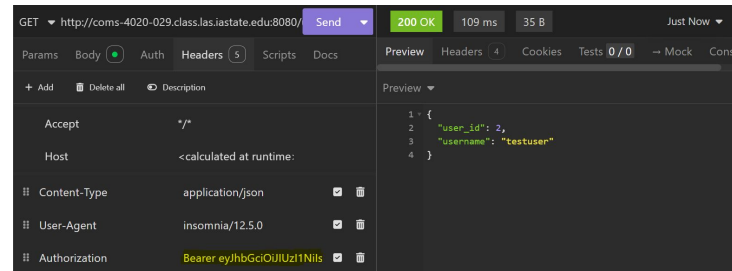
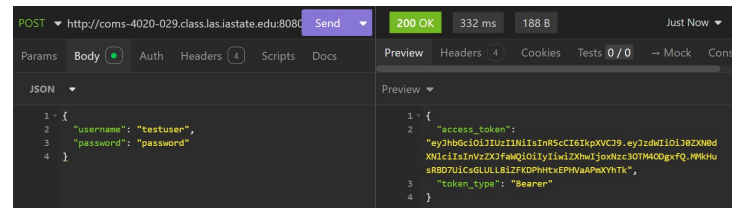
The Portfolios page shows a list of portfolios on the left and a detailed view of 'testyboyo1' on the right. The detailed view includes a balance of \$43185.75 and buttons for Deposit, Withdraw, and Trade. Below this is a table with transaction data.

Ticker	Quantity	Average Cost	Price	Value	Profit
TEST	5	\$100.00	\$42.15	\$210.74	\$-289.26
AAPL	25	\$252.57	\$246.63	\$6165.75	\$-148.50



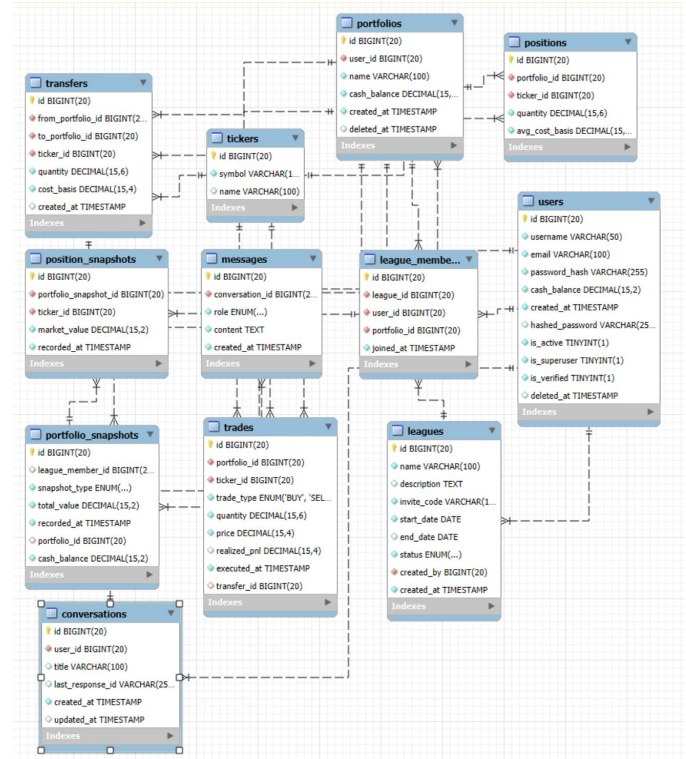
Security + Authentication

- Password hashing
- Login generates Json Web Token (JWT)
 - Gives token
 - Contains user ID, username
 - Expires after 60 minutes
 - Endpoints that are user specific require valid JWT
- Suboptimal: Implemented manually
 - Attempted refactor to 'FastAPI Users' library
 - Custom vs library User class conflict
 - Rebuild of database relations + endpoints
 - No added functionality



Database: stored information

- Lots of relations between stored data
- We store:
 - User information
 - Portfolio information
 - Portfolio snapshots (state at point in time)
 - Held positions
 - Specific information for Stock-AI features
- We do not store:
 - Historical stock pricing
 - Specific stock data (e.g., company description)
 - Why store when available with APIs?



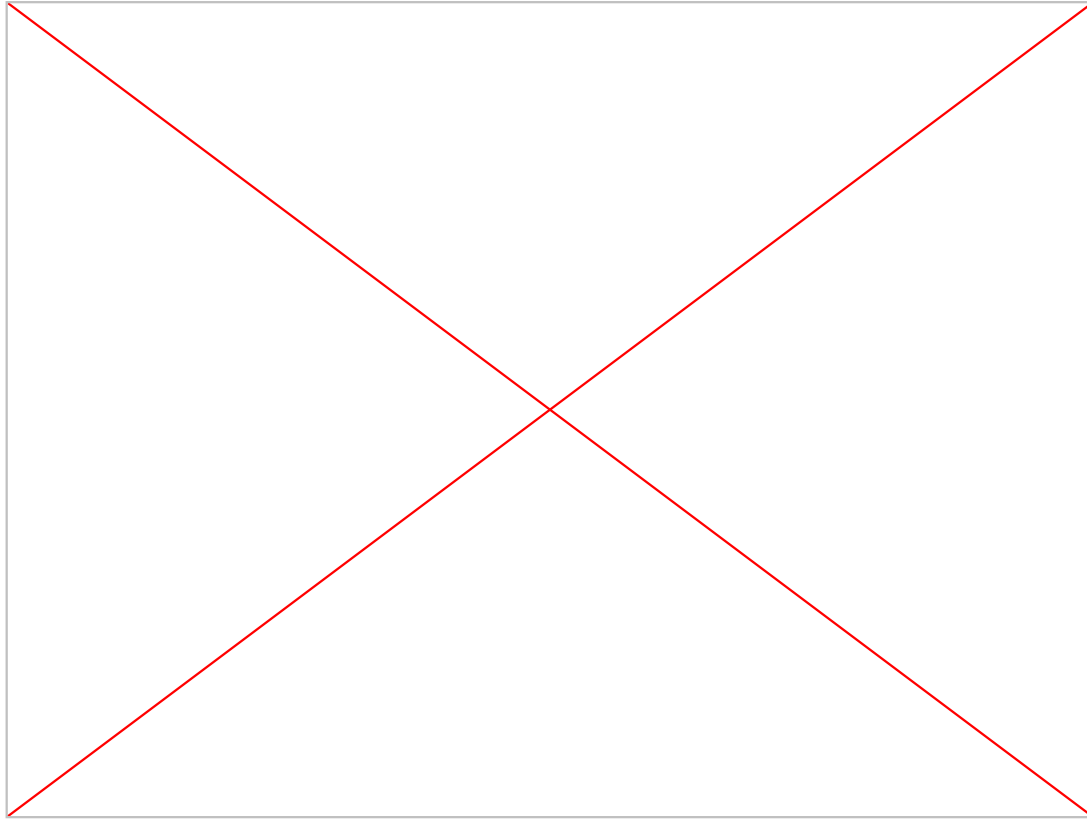
Portfolios, Users: Soft delete

Do not fully remove database entry on deletion

- Add field to mark as deleted
- Preserves related DB entries
- Makes recovery possible

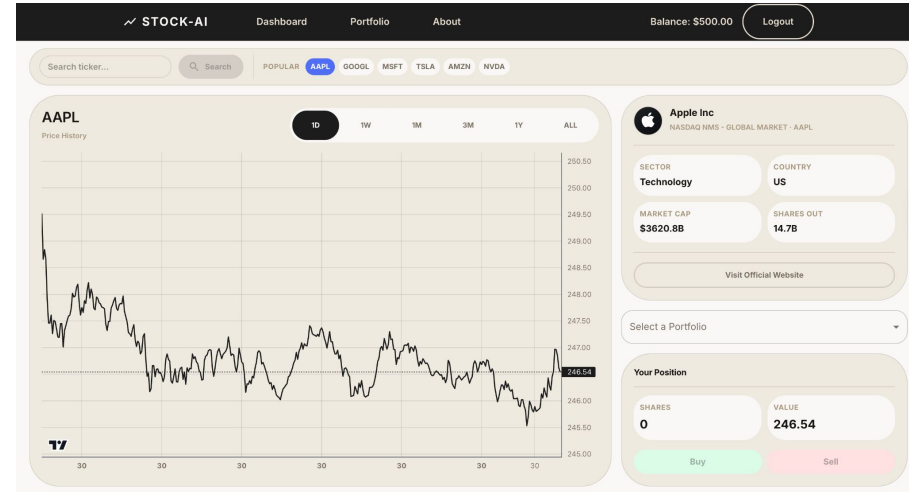
	id	user_id	name	cash_balance	created_at	deleted_at
▶	1	1	Investments Main	877.30	2026-02-24 19:08:36	NULL
	2	1	RenameTest	0.00	2026-02-24 19:08:41	2026-04-14 02:36:09

Users Demo



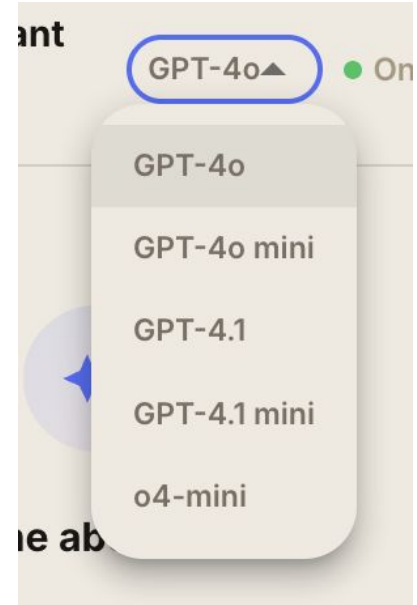
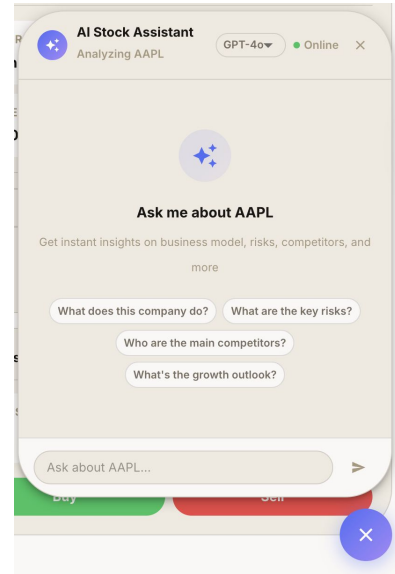
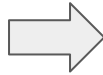
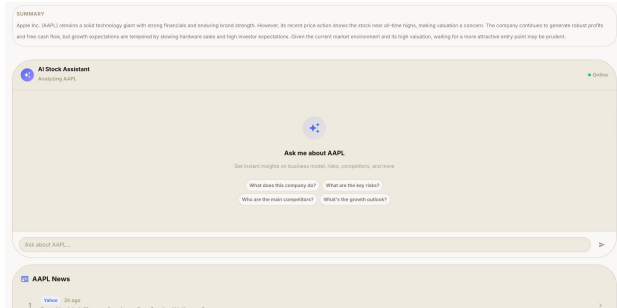
Stock Dashboard

- Search for any stock
- Visualize stock data
- Company information
- Market news



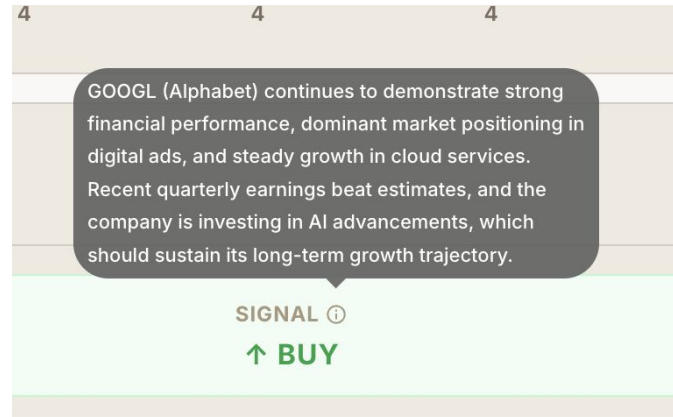
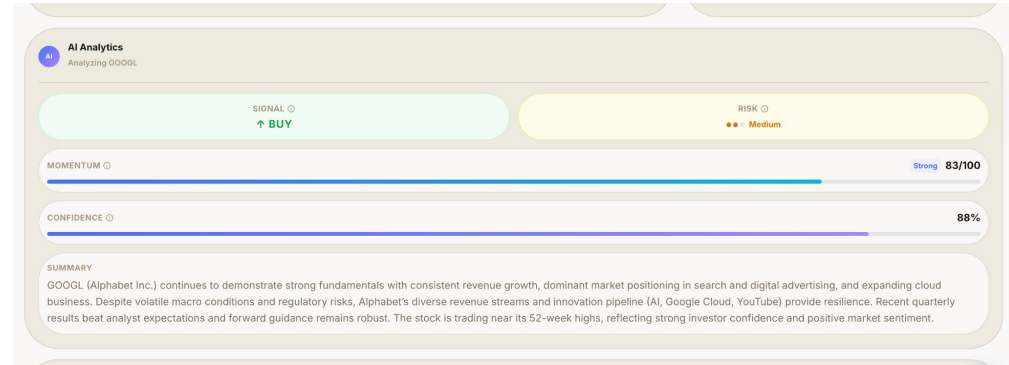
AI Chat Assistant

- Remembers conversation history
- Prompt engineering to keep on track
- Model selection (OpenAI)

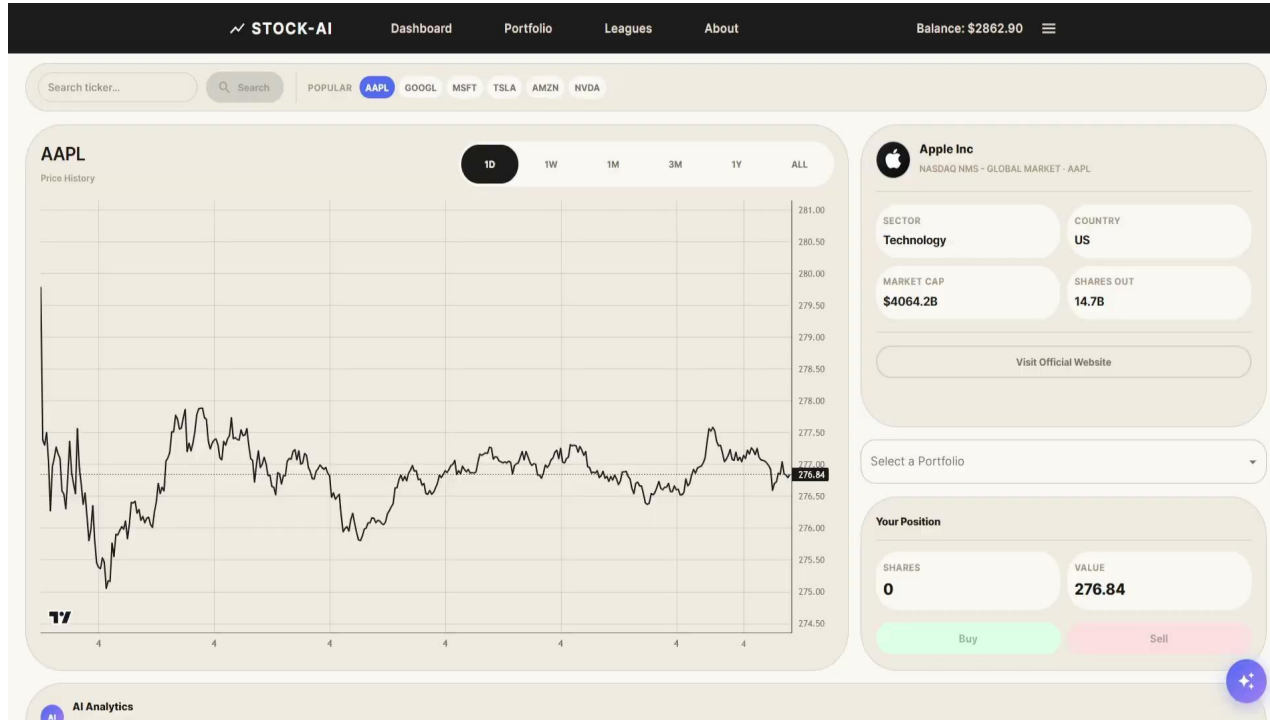


AI Trading Recommendations

- Buy/Sell signal
- Risk analysis
- Summaries
- Explanations



Dashboard Demo



User Portfolios

- User balance & Portfolio balance handled separately
- Organization of owned positions
- Trading done on portfolio basis
- Can transfer positions between portfolios
- Soft deletion

STOCK-AI Dashboard Portfolio Leagues About Balance: \$1362.90

Portfolios

Create Portfolio

Tech Investments
\$5033.77

Misc Investments
\$1500.00

Rename
Delete

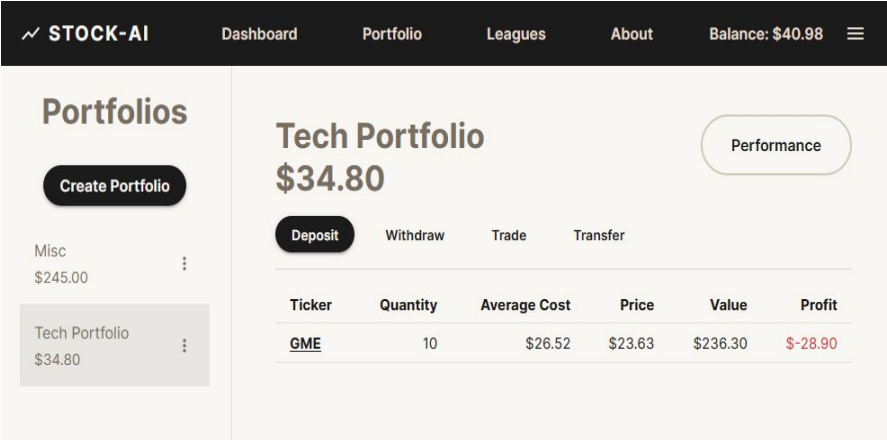
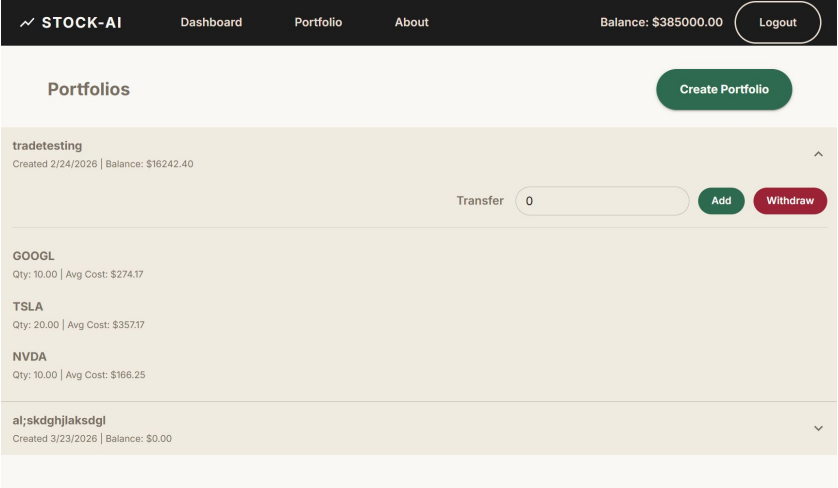
Tech Investments \$5033.77

Performance

Deposit Withdraw Trade Transfer

Ticker	Quantity	Average Cost	Price	Value	Profit
HPE	30	\$28.73	\$29.46	\$883.95	+\$22.11
MSFT	3	\$413.71	\$410.82	\$1232.46	-\$8.67

User Portfolios Design Improvements



User Portfolios Demo

STOCK-AI

DashboardPortfolioLeaguesAbout

Balance: \$4517.74

Portfolios

Create Portfolio

Tech Investments
\$4189.41

Tech Investments
\$4189.41

Tech Investments
\$4189.41

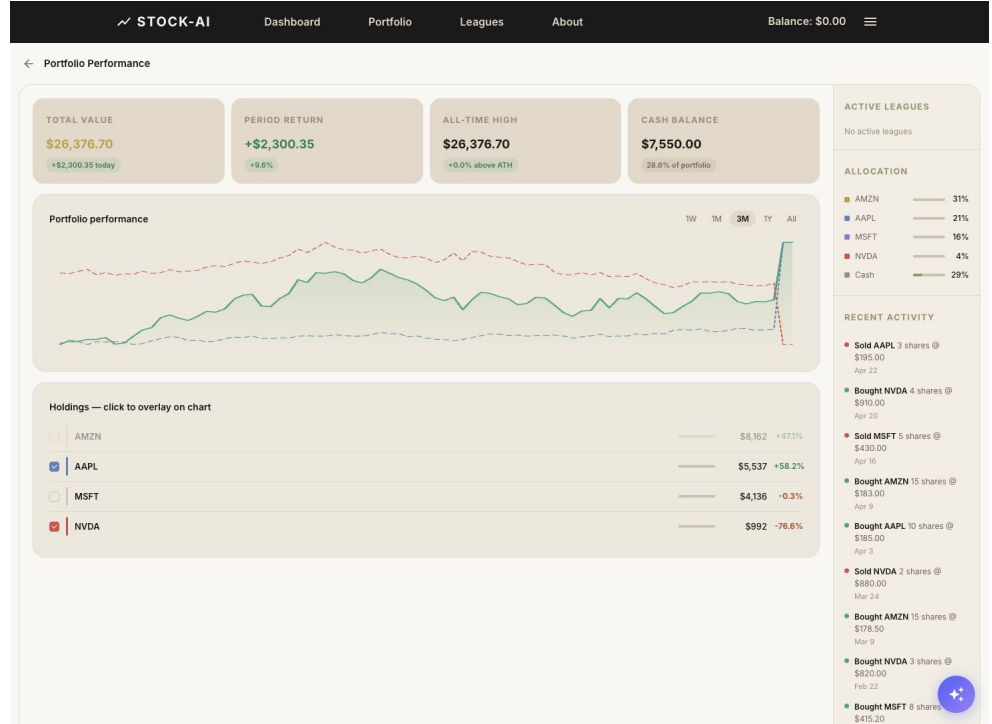
Performance

DepositWithdrawTradeTransfer

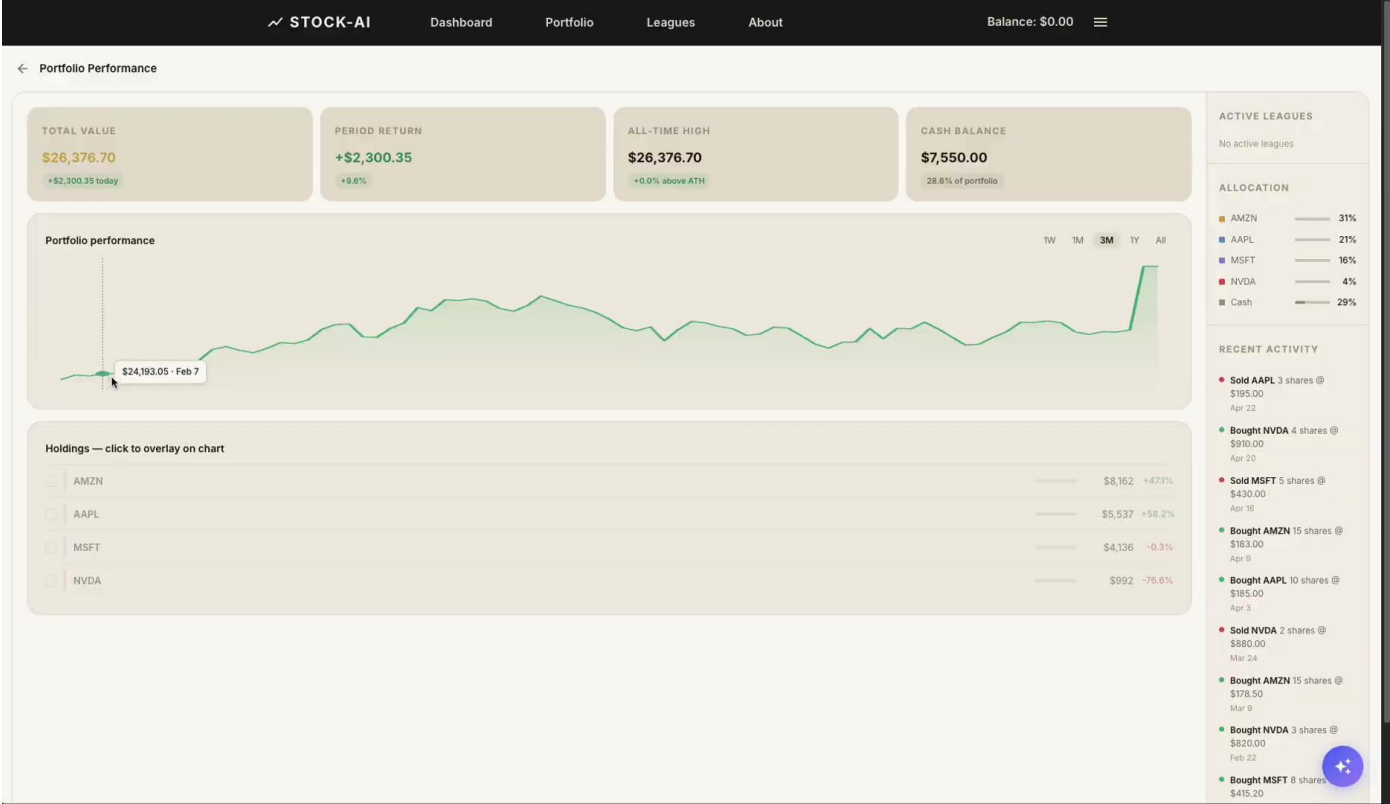
Ticker	Quantity	Average Cost	Price	Value	Profit
HPE	45	\$28.73	\$28.71	\$1291.95	\$-0.90

User Portfolios - Performance

- Value snapshot daily
- Trade history (bottom right)
- Stock allocation breakdown

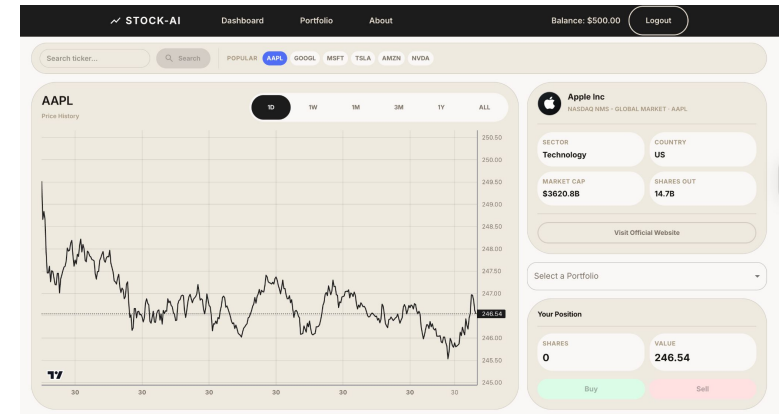
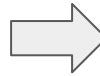
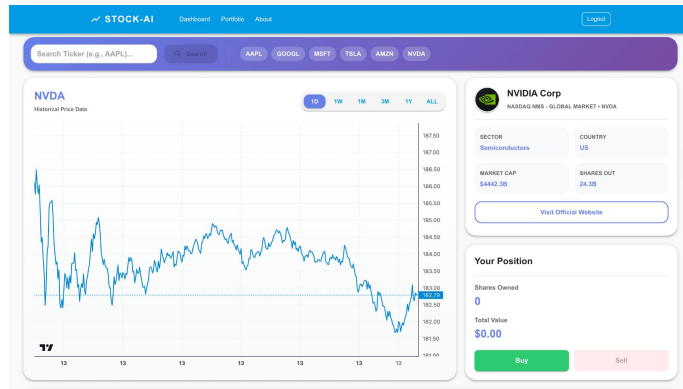
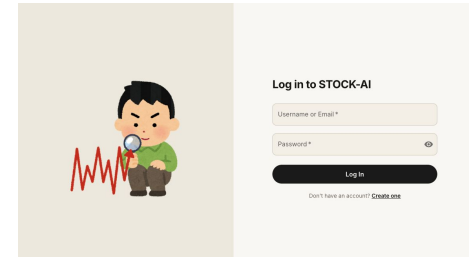


Portfolio Performance Demo



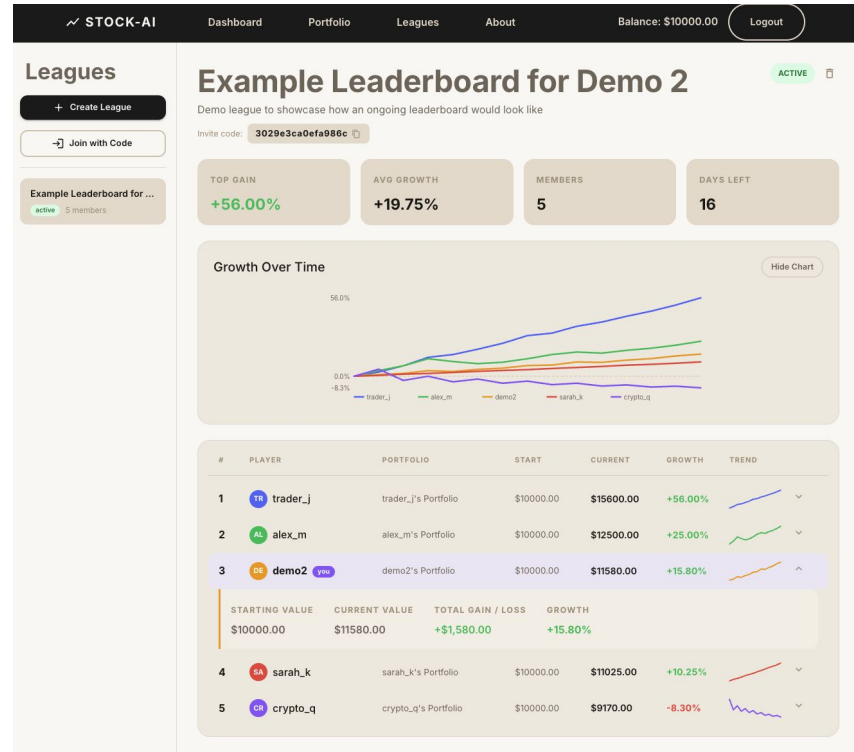
UI Design

- Originally bland, high contrast
- Moved to calm color palette
- Clipart



Competitive Leagues

- Completely optional
- Boost engagement



Competitive Leagues Demo - Expandable Rows + Deletion

STOCK-AI

DashboardPortfolioLeaguesAbout

Balance: \$7500.00

+ Create League

Join with Code

Example Leaderboard for ...
ended 5 members

Friends
ended 1 member

Demo 2
ended 3 members

testvideo
ended 2 members

Example Leaderboard for Demo 2

Demo league to showcase how an ongoing leaderboard would look like

Invite code: 3029e3c0defa986c

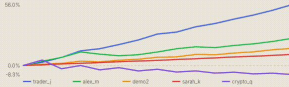
TOP GAIN
+56.00%

AVG GROWTH
+19.75%

MEMBERS
5

DAYS LEFT
0

Growth Over Time



Hide Chart

#	PLAYER	PORTFOLIO	START	CURRENT	GROWTH	TREND
1	trader_j	trader_j's Portfolio	\$10000.00	\$15600.00	+56.00%	
2	alex_m	alex_m's Portfolio	\$10000.00	\$12500.00	+25.00%	
3	demo2	demo2's Portfolio	\$10000.00	\$11580.00	+15.80%	
4	sarah_k	sarah_k's Portfolio	\$10000.00	\$11025.00	+10.25%	
5	crypto_q	crypto_q's Portfolio	\$10000.00	\$9170.00	-8.30%	

STOCK-AI

DashboardPortfolioLeaguesAbout

Balance: \$7500.00

+ Create League

Join with Code

demo league
pending 1 member

Example Leaderboard for ...
ended 5 members

Friends
ended 1 member

Demo 2
ended 5 members

testvideo
ended 2 members

Leagues

This is a league for my friends

Invite code: b8c6fc414b95a908

TOP GAIN
+0.00%

AVG GROWTH
+0.00%

MEMBERS
1

DAYS LEFT
0

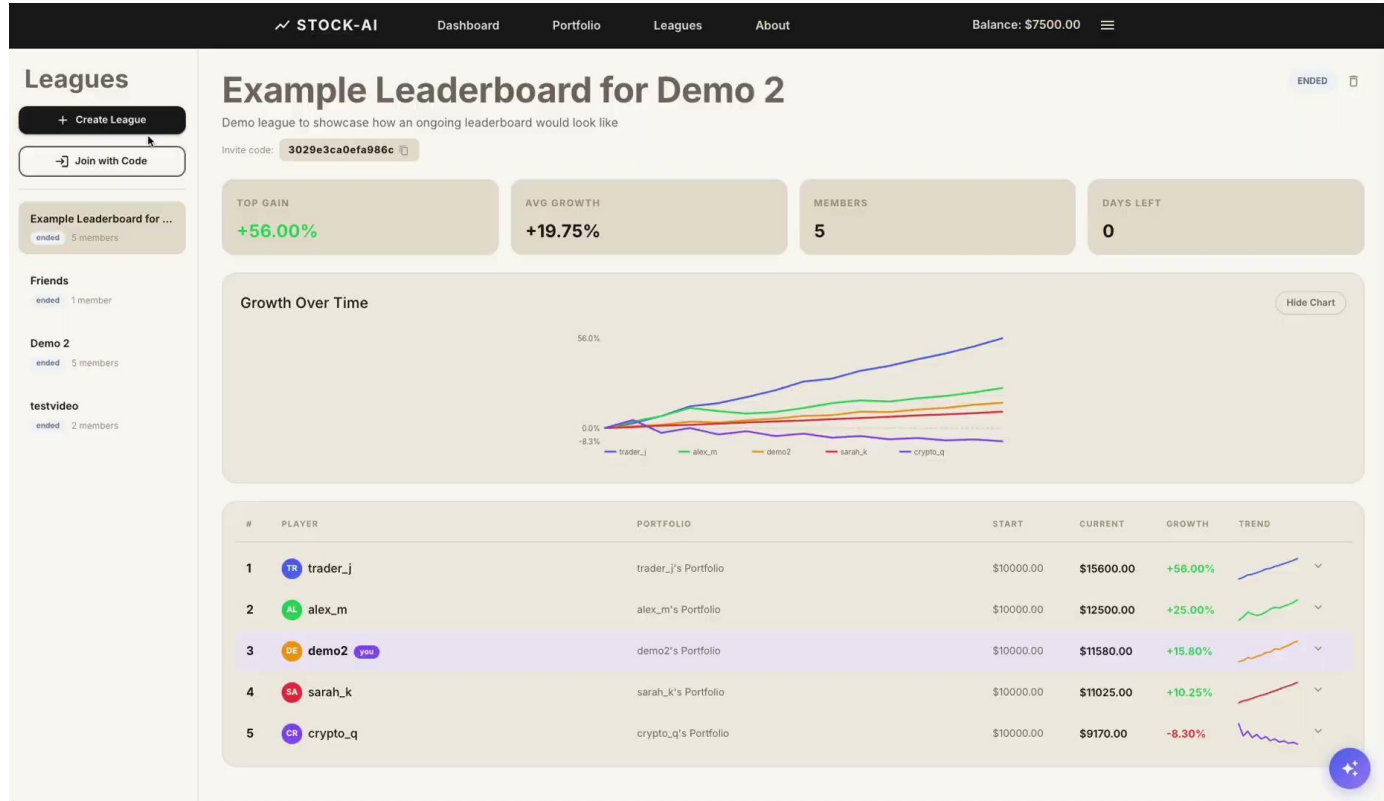
#	PLAYER	PORTFOLIO	START	CURRENT	GROWTH	TREND
No entries yet.						

Friends

Demo 2

testvideo

League Demo - Creation



League Demo - User Joining

STOCK-AI

DashboardPortfolioLeaguesAbout

Balance: \$0.00

Leagues

+ Create League

Join with Code

testvideo
ended 2 members

testvideo

testttttt

Invite code: a7be9aece469a998

TOP GAIN
+0.00%

AVG GROWTH
+0.00%

MEMBERS
2

DAYS LEFT
0

#

PLAYER

PORTFOLIO

START

CURRENT

GROWTH

TREND

No entries yet.

Development challenges

- Deployment + Connectivity
- UI Design
- AI Confusion
- Should have started with Users library

Summary of Contributions

- Andrew Butler

- Backend deployment
- Users + Auth backend
- Portfolios backend
- Stock trading backend

- Ethan Roe

- Users + Auth frontend
- Portfolios frontend
- Stock trading frontend

- Matthew He

- About page
- Leagues
- Portfolio performance
- Daily snapshots (cron job)

- Aaron Trelstad

- Dashboard
- AI Chatbot
- AI Analysis